Regret Is Weighted Forgetting

Michael Timothy Bennett
m@michaeltimothybennett.com

Abstract

How much of a reinforcement learning agent's regret comes from its representation, and how much from its policy? Assume a representation M that compresses histories. It clusters histories into cells, discarding whatever set them apart. Suppose a policy must answer questions about histories, and we weight history-question pairs by question frequency and history importance. A policy can give only one answer per cell, so the merging of histories incurs an irreducible loss of information. Any policy's regret splits into this irreducible loss, and reducible error from bad decisions, so minimising regret means choosing what to represent, not only how to act. I prove its least normalised regret equals $K_\rho(M)$, weight deleted to leave one correct answer per cell. On a stripped-back version of this task, $K_\rho(M)$ is freedom destroyed, where freedom measures how many further commitments a policy leaves open while staying correct. This ranks representations, and chance of generalising to unseen demands is subsequently $e^{-K_\rho(M)}$. I run Proximal Policy Optimization experiments to test this, in which a K_ρ selector uses unlabelled target histories to choose among memory representations. It reduced normalised regret by 8.4%, without using target labels or rewards before selection, in comparison to control selections made by an impurity selector, which gives every within-cell answer conflict equal weight. It closes 95% of the gap between an oracle with target labels and random selection, while the impurity control closes only 63%. This establishes a formal bridge between reinforcement learning and Stack Theory, a formalism of embodied generalisation-optimal learning, with a demonstrated practical use case.

Keywords: representation learning, reinforcement learning, regret, freedom, Stack Theory

1 Introduction

When an agent compresses its observation history into a finite representation, it loses information. Some of that loss is harmless, but the rest of it may not be. Harmful loss can render the correct action ambiguous among histories the representation can no longer tell apart, creating irreducible regret that no policy using that representation can fix. This raises a question; how much irreducible regret does a given representation incur?

Here I prove a decomposition, to resolve this question. Fix a representation M, a function that maps each history h to a state $M(h)$, and fix a finite evaluation distribution over pairs (h, T) of a history and a binary diagnostic test. Sort the pairs into *cells*, where two pairs share a cell when the representation assigns their histories the same state and the same test is being asked. A policy that observes only $M(h)$ and T must answer identically across each cell, so a cell containing pairs with different correct answers forces mistakes. I prove that the minimum average normalised regret over all M-based policies equals the margin-weighted forgetting cost $K_\rho(M)$, the smallest weighted mass of evaluation pairs whose deletion leaves every cell with a single correct answer

(Theorem 3.10).[1] For any actual M-based policy, total regret decomposes into $K_\rho(M)$ plus an avoidable within-cell misreporting term (Theorem 3.14). This is a representation-quality diagnostic analogous to the bias–variance decomposition in supervised learning. It tells you which part of the error is structural and which part is fixable by better optimisation. The experiment in Section 6 tests the resulting representation-search rule on source-trained recurrent Proximal Policy Optimization (PPO) encoders under a target delay shift. A *world* is one sampled delayed-cue environment. Two 24-world replications retain selector agreements. Two more retain stable disagreements, where each choice survives omission of any one of eight target-history folds. Hidden target classes, posterior class probabilities, correct answers, rewards, and regret remain sealed until all source-based choices and policies are written. The K_ρ selector ranks candidates better in the first pair and selects lower-regret representations in the second. Earlier identity checks and diagnostic studies are reported in the electronic supplementary material (ESM).

The decomposition also extends regret minimisation beyond policy selection. For any declared family $\mathcal{M}$ of admissible representations, the least attainable regret is $\inf_{M\in\mathcal{M}} K_\rho(M)$. Policy optimisation removes avoidable within-cell error. Representation search changes the irreducible term (Theorem 3.15).

Stack Theory is a formal framework for *embedded, embodied* and *enactive* generalisation-optimal learning in which a policy is scored by its *freedom*[2], which is the number of ways it can be completed without contradiction, and the central theorems prove that selecting the freedom-maximising correct policy maximises the probability of generalising to unseen demands [6, 11, 15]. This paper is not proposing freedom as a measure, merely translating it to a new context. The optimality claims around freedom are covered at length elsewhere and are beyond the scope of this paper [11]. Recent related work proves invariance of freedom under reparameterisation (in contrast to other measures like flatness), and shows freedom is strongly correlated with accuracy over a variety of benchmarks [12]. Earlier inductive logic programming experiments compared freedom selection with description-length selection and found significant improvements in generalisation rate [6].

Section 4 builds out the relevant vocabulary. I prove that $K_\rho(M)$ is a deficit in a weighted form of freedom on a diagnostic task lifted from the evaluation support (Theorem 4.10).

Note that I said *lifted*. In the general Stack-Theoretic setting, Bayes error depends only on the two weighted label masses inside each cell, whereas freedom depends on extension structure in the embodied language. The equality holds here because the *lifted* diagnostic task deliberately removes that extra structure and treats each cell as an independent binary retain/delete choice. The weights themselves are not new on either side of this research genealogy. The margin weight that converts mistakes into normalised regret in Nayebi's selection-theorem framework [36] and the prior weight native to Stack Theory's generalisation-probability theorems [11] turn out to be the same number. What *is* new is the lift-specific identification. It connects a reinforcement-learning (RL) diagnostic to a programme that links generalisation-optimal learning to selective memory, intervention-sensitive causal structure, free energy, and multiscale biological organisation [9, 11]. For reinforcement learning, this gives a reason to compare learned memories by the decision cost of distinctions they merge, rather than by source return or unweighted cell impurity. Section 6 uses that comparison to select recurrent representations under target delay shift. The broader bridge also connects reinforcement learning [36], optimal embodied learning theorems [13], multiscale

[1]Throughout, "forgetting" means deleting evaluation points from a cell until the optimal bet on that cell is single-valued. It is *not* catastrophic or continual-learning forgetting. $K_\rho(M)$ is the margin-weighted amount of such deletion that the representation forces.

[2]I use *freedom* throughout this paper as it is more semantically accurate, but earlier work also called the same quantity *weakness*, which is why I retain the notation w and the relation $<_w$. The newer terminology foregrounds the quantity's least-commitment interpretation while keeping the historical name explicit [12, 14].

competency architectures [9, 32], liquid and solid brains [47, 48], consciousness [11], and causal world-model literature [36, 43, 44, 50]. The Stack-Theoretic machinery used by the main identity is not a contribution of this paper. Its constructions and optimality theorems were in place [8, 10, 11, 15]. The electronic supplementary material restates and proves each Stack-Theoretic result imported by the bridge. The neural and expected-free-energy papers serve as external comparisons and are cited rather than reproduced. My contribution in this paper paper is to the mathematical core of that programme.

A scope note on the term *regret.* The identity below concerns fixed-distribution normalised betting regret in the sense of [36], namely $\delta_i = 1 - V_i^\pi / V_i^\star$ averaged over a finite evaluation distribution over history-test pairs. It is not cumulative regret, Markov decision process (MDP) value loss, or worst-case minimax regret. The bridge transports to those settings only in special cases. Keeping this scope explicit is what allows the cell-by-cell decomposition to be exact rather than up-to-constants.

The identity is also a proof technique, and the contributions are as follows.

1. An exact identity, stated and proved in self-contained reinforcement-learning terms, equating the minimum average normalised regret over M-based policies with the margin-weighted forgetting cost $K_\rho(M)$ (Theorem 3.10), together with a policy-wise decomposition of any actual policy's regret into irreducible aliasing cost plus avoidable misreporting (Theorem 3.14).

2. Representation search and corollaries in pure RL language. Minimising over any declared representation family reduces to minimising $K_\rho(M)$ (Theorem 3.15). The same reduction yields a regret-based preorder on abstractions (Theorem 5.1), a diagnostic-quotient convergence theorem (Theorem 5.2), Lipschitz stability under margin estimation error (Theorem 5.3), a multi-class generalisation (Theorem 3.18), and a measurable version in the electronic supplementary material.

3. The bridge, together with its limits. A proof that $K_\rho(M)$ equals the weighted freedom deficit $Z - W^\star(M)$ on a lifted diagnostic task (Theorem 4.10), with the RL margin weight and the Stack-Theoretic prior weight identified as the same quantity on that lift. A proof that the equality is confined to that lift, since the freedom deficit dominates the partition Bayes error under any admissible family and the inequality turns strict once cross-cell extension constraints bind (Theorem 4.11), together with a four-point witness for strictness (Theorem 4.12).

4. A cross-framework consequence via the bridge. Under the canonical independent prior, the optimal M-based policy generalises with probability $e^{-K_\rho(M)}$ (Theorem 5.4). External work fixes the scope of any expected-free-energy interpretation [14].

5. A recurrent PPO representation-search experiment with two 24-world replications retaining selector agreements and two 24-world replications restricted to stable selector disagreements. An estimate of K_ρ, calibrated from labelled source episodes, uses unlabelled target histories to rank and select fixed-budget memories before target labels or rewards are revealed (Section 6).

How to read this paper, and what I assume. No prior familiarity with Stack Theory is assumed. The core identity is fully self-contained in Section 3 and can be read on its own as a representation result in standard reinforcement-learning terms. Section 6 then applies the representation-search corollary to learned recurrent states. The Stack-Theoretic reading is optional. Section 4 builds every notion it needs from scratch with a running example, while Sections 4.5 and 5 develop the bridge. Earlier two-sided verification and diagnostic experiments are in the electronic supplementary material. Every Stack-Theoretic result used to prove the bridge is restated and *proved* in the electronic supplementary material. Related neural and expected-free-energy results are cited as supporting companion works, but are otherwise beyond the scope of this paper. The electronic supplementary material records what each result rests on so any single claim can be checked by following a short chain of references rather than reading the whole paper.

The rest of the paper is organised as follows. Section 2 discusses related work. Section 3 sets up the diagnostic problem and proves the exact reduction and policy-wise decomposition, entirely in reinforcement-learning terms. Section 4 introduces the Stack-Theoretic notions used later, and Section 4.5 states the lifted bridge in that language. Section 5 derives new results via the bridge. Section 6 gives a recurrent PPO representation-search experiment under target delay shift. The electronic supplementary material gives full methods, audit results, earlier diagnostic experiments, and comparisons with external freedom evidence.

2 Related work

2.1 Selection theorems and genealogy

The broad claim that competence pressure forces agents to acquire internal structure has several independent lines. The Stack-Theoretic line proves freedom optimality under the maximally uninformative extension model (2023) and derives intervention-sensitive causal identities under explicit preconditions (2023–2025) [6, 7, 11, 15]. Richens and Everitt derive robust-causal-model results under distributional shift (2024) [43]. Nayebi works in a standard partially observable Markov decision process (POMDP) setting and derives quantitative selection theorems for predictive state, memory, modularity, regime tracking, and recoding match (2026) [36]. These five structural selection theorems are distinctive contributions of Nayebi's programme that do not have direct Stack-Theoretic antecedents. The two frameworks also handle noise differently. Rabin and Scott showed that the languages recognisable by deterministic and non-deterministic finite automata coincide [42]. In the Stack-Theoretic framework, the role played by stochastic policies in Nayebi's setting is played by selective forgetting, which handles noise and inconsistency by discarding outlying data rather than by randomising over actions [7, 11]. A detailed comparison is in Table 1 and a claim-by-claim antecedent inventory is in the electronic supplementary material. The regret identity, the policy-wise decomposition, the multi-class generalisation, the measurable version, and the derived consequences in Section 5 are new to this paper.

The representational results can be situated within a broader debate, where Brooks pushed intelligence without explicit internal representation, and Marr and later work in representation learning and transfer emphasise structured internal states [4, 16, 31, 54]. The causal representation learning programme argues that disentangled causal variables are the natural targets [5, 23, 28, 45]. Nayebi's recoding theorem and Cao and Yamins' contravariance principle are best read against that background [17, 36].

2.2 State abstraction and bisimulation

The representation-test quotient in my bridge theorem is a form of state abstraction. The state abstraction literature in RL is mature and gives important context. Li, Walsh, and Littman give a unified treatment of five abstraction schemes for MDPs, ranging from model-irrelevance to π^*-irrelevance and bisimulation, and analyse which preserve optimal planning [30]. Givan, Dean, and Greig earlier studied equivalence notions and model minimisation for MDPs, identifying bisimulation as the finest equivalence of practical interest [22]. Ferns, Panangaden, and Precup introduced

Table 1: How the three lines relate.

	Bennett	**Richens & Everitt**	**Nayebi**
Main setting	Task extensions and policy freedom	Causal models under distributional shifts	POMDP betting tasks and predictive tests
Competence variable	Freedom, with selective forgetting for noise	Robust regret-bounded adaptation	Average-case normalised regret
Core structural conclusion	Optimal generalisation selects policies with greater freedom and, under preconditions, intervention-sensitive causal identities	Robust agents must learn an approximate causal model	Low regret selects predictive state, memory, modularity, regime tracking, and recoding match

quantitative bisimulation metrics that measure state similarity continuously rather than through hard partitions, providing value-function bounds on the cost of aggregation [19, 20]. Abel and collaborators extended these ideas to approximate and lifelong settings [1, 2].

My quotient cells differ from bisimulation cells in an important respect. Bisimulation groups states by identical transition and reward structure, while my representation-test cells group history-test points by what the representation M can see and which diagnostic test is applied. The bridge theorem then settles how much regret this grouping forces. In that sense it complements the bisimulation programme. Bisimulation metrics bound value-function loss, while my theorem identifies the exact regret cost at a given quotient.

More recent deep RL work brings bisimulation ideas into the function-approximation regime. Zhang et al. learn representations that respect a bisimulation distance [55], Castro scales bisimulation computation to large deterministic MDPs [18], and Gelada et al. learn latent-space models that preserve MDP structure [21]. Nayebi's selection theorems and my bridge result operate at a more abstract level. They characterise when low regret forces a representation to preserve certain distinctions, regardless of how the representation is learned.

Remark 2.1 (Bisimulation distance and cell impurity). In an MDP with known transition and reward structure, if two states s, s' belong to the same representation-test cell C and the bisimulation metric $d_\sim(s, s')$ of Ferns, Panangaden, and Precup [19] is large, then s and s' are likely to harbour different optimal actions, so their cell will be impure and will contribute positively to $K_\rho(M)$ when the diagnostic tests are value- or optimal-action diagnostics preserved by that bisimulation relation. Conversely, if every pair in a cell has $d_\sim(s, s') = 0$ and the diagnostic class is preserved by that relation, then the cell is pure and contributes zero. For arbitrary diagnostic tests, bisimulation alone need not determine $p_T(h)$, so the implication requires this diagnostic-class restriction. So $K_\rho(M)$ can be read as an aggregate measure of how much bisimulation-relevant structure the representation has thrown away for the specified diagnostic quotient. The two formulations are complementary. Bisimulation metrics bound value-function loss, while $K_\rho(M)$ gives the exact regret cost at a given quotient.

Additional predictive-state material is in the electronic supplementary material.

2.3 Memory, latent states and world models

Partial observability turns history compression into a central RL problem. Recurrent model-free RL can match or exceed specialised POMDP methods when architecture and optimisation are tuned [37]. Transformers extend usable memory much further, but experiments separating memory from credit assignment find that their advantage concerns memory rather than long-horizon credit assignment [38]. The POPGym benchmark isolates memory-focused POMDPs [35], while Partially Observable Benchmarks in JAX (POBAX), a benchmark suite implemented in the JAX numerical-computing library, broadens evaluation to memory-improvable domains spanning navigation, games and visual control [51].

A second line learns predictive latent states. Self-predictive RL connects state and history objectives through latent prediction [39]. DreamerV3 uses a recurrent state-space model across more than 150 tasks [24], while a second-generation temporal-difference model-predictive-control method, TD-MPC2, learns latent dynamics for single-task and multi-task continuous control across 104 tasks [25]. Richens, Everitt and Abel prove that any agent generalising to multi-step goal-directed tasks must encode a predictive environment model [44]. Action-sufficient goal representations expose a related failure mode. A representation can preserve value information while merging goals that require different actions [26].

This paper addresses a complementary question. Given several source-trained history representations, how much normalised regret does each representation force, and which one should an RL system deploy before target rewards are available? I do not introduce a new recurrent architecture, world model or benchmark state of the art. I give a decision-weighted cost for information merged by a declared representation, then use it as a target-input selector under delay shift. This selector sits between representation learning and deployment. It can compare recurrent checkpoints, encoders or fixed-budget memory quotients without changing their training algorithm.

2.4 Causal inference and the causal hierarchy

The causal side of the genealogy involves more than Richens and Everitt alone. Pearl's structural causal model framework [40, 41] and the Spirtes–Glymour–Scheines algorithmic tradition [49] provide the language in which "learning a causal model" is given exact meaning. Bareinboim et al.'s causal hierarchy theorem formalises the separation between observational, interventional, and counterfactual reasoning [3]. Lattimore, Lattimore, and Reid study causal bandits, where the agent must learn which interventions are effective [29].

Bennett's emergent causality result [7] proved that under freedom maximisation, representations of causal interventions emerge as variables in the embodied language, without presupposing a do-operator. The key argument is that an agent which confuses intervention with passive observation is forced into a needlessly committal policy, one with a smaller extension and therefore stronger in the freedom sense, than an agent which distinguishes them, so the pressure to maximise freedom forces the distinction to emerge. That result, and the subsequent causal identity theorems in the thesis [11], are antecedent to the Richens–Everitt result that robust agents must learn approximate causal models under distributional shift [43].

The two lines are somewhat complementary. Bennett's argument was that since freedom maximisation is the optimal choice of learning heuristic [11], an agent that seeks to be optimal within the constraints of its body must use it. Such agents *must* learn causal models [7, 11]. In contrast, Richens and Everitt claimed *robust* agents must learn causal models, basing their claim on reinforcement learning. These are approximately the same claim derived from two different formalisms, which suggests a bridge between Stack Theory and reinforcement learning may be

$$\begin{array}{l} (\mathcal{X}, \mu, y, \rho) \xrightarrow{\text{quotient by } M} \mathcal{C}_M \xrightarrow{\text{drop the cheaper label in each mixed cell}} K_\rho(M) \\ K_\rho(M) \xrightarrow{\text{lift to a diagnostic Stack task}} Z - W^\star(M). \end{array}$$

Figure 1: The reduction at a glance. Sort the history-test points into the cells induced by M. Drop the cheaper label class in each mixed cell. Read the same number as a weighted freedom deficit on the lifted task (Section 4.5).

beneficial. In this paper, the regret ordering (Theorem 5.1) gives a quantitative counterpart to the qualitative claim that robust agents must learn causal models. If the causal partition refines the diagnostic partition $\mathcal{P}_+^\star$, it sits at zero K_ρ and incurs no irreducible regret. If it does not, the bridge tells you how much regret that costs. Richens and Everitt handle worst-case robustness across distributions. The bridge presented here handles the cost at a fixed evaluation distribution.

3 Regret as weighted forgetting

Here the identity is given in finite form, stated and proved entirely in reinforcement-learning terms. Nothing in this section depends on Stack Theory. A reader who only wants the RL result can read this section and the experiments and stop. The Stack-Theoretic reading of the same number is developed in Sections 4 and 4.5. A measurable version is given in the electronic supplementary material. The finite version is stated first because it keeps the reduction transparent and matches the finite task-extension setting in which freedom was originally defined [6].

3.1 Setup

This subsection defines, in words first and then symbolically, the four objects that the rest of the paper assumes. Readers familiar with Nayebi's betting framework can skip to Theorem 3.4.

Definition 3.1 (History, test, conditional)**.** A *history* h is the agent's observed interaction record up to a decision point. The space of histories is left abstract. Concrete instantiations include observation-action sequences in a POMDP, sensor readings up to a query time, or a statement in the embodied language of Section 4 summarising the body's state. A *test* $T \in \mathcal{T}$ is a binary diagnostic query. Its outcome is a binary random variable $Z_T \in \{0, 1\}$. The *conditional*

$$p_T(h) := \Pr(Z_T = 1 \mid h) \in [0, 1]$$

is the true environment probability that the test fires given the history. The agent does not observe $p_T(h)$ directly. The exact identity below uses $p_T(h)$ as a population quantity. Theorem 5.3 controls what happens when $p_T(h)$ is replaced by an empirical estimate. The multi-class case replaces Z_T by a finite-valued outcome variable and is recovered in Theorem 3.18.

Definition 3.2 (Evaluation distribution)**.** A finite *evaluation distribution* is a finite list of history-test pairs $\{x_i = (h_i, T_i)\}_{i=1}^n$ together with positive weights $\mu_i > 0$ summing to one. The pairs are the support points at which the representation will be diagnosed. The weights say how often each support point is visited under the evaluation regime. They are part of the diagnostic problem, not learned by the agent.

Definition 3.3 (M-based policy)**.** A *representation* M (also called a *memory representation*, since it fixes what the agent retains about the history) is any function on histories. Its observable output at history h is $M(h)$. The *cells* $\mathcal{C}_M$ are the equivalence classes of support points under the relation $i \sim_M j$ iff $M(h_i) = M(h_j)$ and $T_i = T_j$. Two support points share a cell when the representation cannot tell their histories apart and the same test is being asked. An *M-based policy* π chooses one report probability $q_C \in [0, 1]$ for each cell $C \in \mathcal{C}_M$. The policy observes only the cell label, not the underlying history. This is what *representation-based* means in this paper.

The betting protocol is the natural one. On support point $x_i = (h_i, T_i)$ in cell C, the policy bets that the test fires with probability q_C and that it does not with probability $1 - q_C$. The bet succeeds when it matches the realised outcome Z_{T_i}, so the success probability is

$$V_i^\pi = q_C\, p_i + (1 - q_C)(1 - p_i),$$

while an unrestricted policy that observes h_i itself achieves

$$V_i^\star = \max\{p_i,\ 1 - p_i\}.$$

With these objects fixed, the following compact notation is used throughout. Let

$$\mathcal{X} = \{x_1, \ldots, x_n\}, \qquad x_i = (h_i, T_i),$$

with masses $\mu_i > 0$ and $\sum_i \mu_i = 1$. For each x_i, let

$$p_i := p_{T_i}(h_i), \qquad m_i := \left|p_i - \tfrac{1}{2}\right|.$$

Let $y_i \in \{0, 1\}$ denote the optimal bet, where $y_i = 1$ when $p_i \geq \frac{1}{2}$ and $y_i = 0$ otherwise. When $p_i = \frac{1}{2}$, the choice of y_i is arbitrary because the weight below is then zero.

For any policy π, write $\delta_i(\pi) := 1 - V_i^\pi / V_i^\star$ for the normalised regret on support point x_i, with V_i^π and $V_i^\star$ as in Theorem 3.3. The partially observed decision-problem setting follows the standard POMDP and state-abstraction literature [27, 30, 50]. The fixed-distribution normalised-regret diagnostic follows Nayebi [36].

Definition 3.4 (Margin weight)**.** For each support point x_i, define

$$\rho_i := \frac{2m_i}{\frac{1}{2} + m_i} \in [0, 1].$$

Note that $\frac{1}{2} + m_i = \max\{p_i, 1 - p_i\}$, so ρ_i is the normalised regret penalty per unit of wrong-action mass on test i. This is the multiplier that converts wrong-action probability into the normalised regret $\delta_i = 1 - V_i^\pi / V_i^\star$ used in the RL selection-theorem literature [36], and the same multiplier arises as the natural prior weight w_P in Stack Theory (Sections 4 and 4.5) [11]. When a test is nearly a coin flip, ρ_i is near zero. When a test is decisive, ρ_i is large. The role is analogous to the margin-based weighting that appears in information-theoretic accounts of lossy compression. Uninformative distinctions are cheap to lose and informative ones are expensive [52].

Remark 3.5 (Why the bridge is natural)*.* The cells come from fixing the representation M and keeping the test variable visible. The deletion move is selective forgetting on the diagnostic support. The margin weights ρ_i, the cells, and the deletion move all have native Stack-Theoretic antecedents in w_P, selective forgetting, and the diagnostic extension model (see Sections 4 and 4.5) [11]. The prior weight w_P appeared in the Stack-Theoretic generalisation-probability theorems [11]. The margin weight ρ_i appeared later as Nayebi's normalised regret multiplier [36]. The contribution of this paper is the lift-specific identification of these as the same quantity. The cellwise structure that allows for the bridge follows from that identification. Nayebi's betting framework is the RL-side formalisation at the bridge's second endpoint.

Remark 3.6 (Scope of the normalisation). The exact identity depends on normalised regret $\delta_i = 1 - V_i^\pi / V_i^\star$ and the resulting margin weight ρ_i. Under unnormalised regret $V_i^\star - V_i^\pi$ or under alternative loss functions, the cellwise structure survives but the exact identification requires a different weight. Specifically, unnormalised regret replaces ρ_i with the weight $2m_i$, giving a different margin-weighted deletion cost on the diagnostic support. The bridge to weighted freedom (Theorem 4.10) requires the normalised form because it matches the prior-weighted quantity w_P native to Stack Theory.

Lemma 3.7 (Pointwise regret decomposition). *Let π be any M-based policy, and let q_C be its report probability on cell C. For $i \in C$,*

$$\delta_i(\pi) = \rho_i\Big((1 - q_C)\mathbf{1}\{y_i = 1\} + q_C\mathbf{1}\{y_i = 0\}\Big).$$

Proof. If $y_i = 1$, then $p_i \geq \frac{1}{2}$ and the optimal report is $q_C = 1$. The success probability under report probability q_C is

$$V_i^\pi = q_C p_i + (1 - q_C)(1 - p_i),$$

while the optimal success probability is

$$V_i^\star = p_i = \tfrac{1}{2} + m_i.$$

Hence

$$\delta_i(\pi) = 1 - \frac{V_i^\pi}{V_i^\star} = \frac{2m_i(1 - q_C)}{\frac{1}{2} + m_i} = \rho_i(1 - q_C).$$

If $y_i = 0$, the symmetric calculation gives $\delta_i(\pi) = \rho_i q_C$. □

Interpretation. Regret is just mistake probability scaled by how informative the test is. That scaling is the only reason the bridge needs a weighted, rather than raw, forgetting variable.

Definition 3.8 (Margin-weighted forgetting cost). The margin-weighted forgetting cost[3] of M is

$$K_\rho(M) := \min\Big\{\sum_{i \in B} \mu_i \rho_i \;\Big|\; B \subseteq \{1, \ldots, n\}, \text{ and for every } C \in \mathcal{C}_M, \; y \text{ is constant on } C \setminus B\Big\}.$$

The cost of forgetting point i is its evaluation mass times its margin weight. You forget just enough so that, inside every representation-test cell, the correct bet becomes single-valued. This is a weighted version of the selective deletion that appears in the state abstraction literature when one asks how much structure a given partition can preserve [22, 30].

Before stating the theorem, here is a minimal worked example.

Example 3.9 (A two-cell warm-up). Take three support points x_1, x_2, x_3 under a single test T, with $\mu_i = 1/3$ each and conditional probabilities $p_1 = 0.9$, $p_2 = 0.1$, $p_3 = 0.7$. The optimal bets are $y_1 = 1$, $y_2 = 0$, $y_3 = 1$. The margin weights are $\rho_1 = 8/9$, $\rho_2 = 8/9$, $\rho_3 = 4/7$.

[3] The word *forgetting* here refers to selective deletion of support points to purify representation cells, in the sense used in the thesis [11]. It is not catastrophic forgetting in the continual-learning sense (loss of past task performance under sequential training) and does not refer to weight overwriting in neural networks.

Suppose representation M collapses h_1 and h_2 to the same code while keeping h_3 separate. Then $\mathcal{C}_M$ has two cells, $C_{12} = \{x_1, x_2\}$ and $C_3 = \{x_3\}$. On C_{12}, $A_{C_{12}} = \mu_1\rho_1 = 8/27$ (the mass on $y_i = 1$) and $B_{C_{12}} = \mu_2\rho_2 = 8/27$ (the mass on $y_i = 0$). The minimum is $8/27$, so this cell forces a K_ρ contribution of $8/27$ regardless of which label class the policy commits to. The cell is balanced. On C_3, $A_{C_3} = \mu_3\rho_3 = 4/21$ and $B_{C_3} = 0$, so the minimum is zero. Total $K_\rho(M) = 8/27 \approx 0.296$.

A different representation M' that kept h_1 separate from h_2 would induce three pure cells, giving $K_\rho(M') = 0$. So aliasing h_1 and h_2 costs $8/27$ in irreducible regret. The theorem below establishes that the same picture works for arbitrary representations, that the cellwise rule is a theorem rather than a heuristic, and that the same number equals the minimum margin-weighted deletion needed to purify the cells.

Theorem 3.10 (Exact reduction). *For every fixed memory representation M,*

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \, \delta_i(\pi) = K_\rho(M).$$

Proof. For each cell $C \in \mathcal{C}_M$, define

$$A_C := \sum_{i \in C,\ y_i = 1} \mu_i \rho_i, \qquad B_C := \sum_{i \in C,\ y_i = 0} \mu_i \rho_i.$$

By Theorem 3.7, any M-based policy with report probability q_C on cell C contributes

$$A_C(1 - q_C) + B_C q_C$$

to expected regret on that cell. This is affine in q_C, so its minimum over $q_C \in [0, 1]$ is

$$\min\{A_C, B_C\}.$$

Summing over cells gives

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \, \delta_i(\pi) = \sum_{C \in \mathcal{C}_M} \min\{A_C, B_C\}.$$

Now compute the forgetting cost. To make y single-valued on a fixed cell C, one must delete either every $y_i = 1$ point in C or every $y_i = 0$ point in C. Deleting anything less leaves both labels in the cell. So the minimum forgetting cost on cell C is

$$\min\{A_C, B_C\}.$$

Summing over cells gives

$$K_\rho(M) = \sum_{C \in \mathcal{C}_M} \min\{A_C, B_C\}.$$

Comparing the two displays proves the theorem. □

Remark 3.11 (Computational cost). Despite the combinatorial phrasing of Theorem 3.8, $K_\rho(M)$ is computable in $O(n)$ time once the representation-test cells are formed, because the minimum deletion on each cell decomposes independently into $\min\{A_C, B_C\}$. No search over subsets is required.

Interpretation. Once the task is quotiented by representation-test cell, regret is the cheapest weighted deletion needed to make the correct action a function of that cell. The two quantities are equal, not merely analogous.

Remark 3.12 (The optimal bet is selective forgetting). The identification is not merely numerical. The optimal M-based policy's per-cell binary choice (bet on $y = 1$ or $y = 0$) is itself a deletion operator. In each cell, the policy commits to one label class and discards the other. The discarded class is the cheaper one, and its margin-weighted mass is the cell's contribution to $K_\rho(M)$. So the RL-side optimal policy is *performing* selective forgetting on the diagnostic support, rather than merely computing a cost equal to it.

Remark 3.13 (Relation to weighted Bayes error). The cellwise quantity $\min\{A_C, B_C\}$ is recognisable as a weighted Bayes error on the representation-test partition. From a statistical decision theory perspective, this is the conditional Bayes risk of the partition under a margin-weighted 0-1 loss. So $K_\rho(M) = \sum_C \min\{A_C, B_C\}$ is a weighted Bayes error on that partition. The contribution is the three-way identification on the lifted diagnostic task. It equals Nayebi's normalised regret exactly, rather than merely up to bounds. It equals the minimum margin-weighted deletion cost in the selective-forgetting sense. And, after the finite support is lifted into the structureless diagnostic task of Section 4.5, it equals a weighted freedom deficit (Theorem 4.10). The last clause is deliberately scoped. The lift gives each support point one unseen output and treats the admissible choices as independent across cells, so weighted freedom reduces to retained weighted mass. On a task with non-trivial extension structure, freedom is not a Bayes error. It also depends on cross-cell extension constraints, and the deficit only upper-bounds the partition Bayes error as described in Theorem 4.11. The first equality gives RL semantics, the second gives a forgetting and compression interpretation, and the third connects the lifted diagnostic task to the generalisation-optimal learning programme.

Corollary 3.14 (Policy-wise decomposition)**.** *For each cell $C \in \mathcal{C}_M$, choose any*

$$q_C^\star \in \arg\min_{q\in[0,1]} \big(A_C(1-q) + B_C q\big).$$

Then every M-based policy π satisfies

$$\sum_{i=1}^{n} \mu_i\, \delta_i(\pi) = K_\rho(M) + \sum_{C\in\mathcal{C}_M} |A_C - B_C|\, |q_C - q_C^\star|.$$

In particular,

$$K_\rho(M) \le \sum_{i=1}^{n} \mu_i\, \delta_i(\pi),$$

with equality when the policy is cellwise optimal on every non-tied cell.

Proof. If $A_C > B_C$, then the minimum is attained at $q_C^\star = 1$ and

$$\begin{aligned} A_C(1-q_C) + B_C q_C &= B_C + (A_C - B_C)(1-q_C) \\ &= \min\{A_C, B_C\} + |A_C - B_C|\, |q_C - q_C^\star|. \end{aligned}$$

If $A_C < B_C$, the minimum is attained at $q_C^\star = 0$ and the same identity becomes

$$\begin{aligned} A_C(1-q_C) + B_C q_C &= A_C + (B_C - A_C) q_C \\ &= \min\{A_C, B_C\} + |A_C - B_C|\, |q_C - q_C^\star|. \end{aligned}$$

If $A_C = B_C$, both sides reduce to A_C for every q_C. Summing over cells and using Theorem 3.10 proves the result. $\square$

Interpretation. This separates two kinds of regret. The first part is representation cost. It is the aliasing penalty you cannot remove without changing M. The second part is execution cost. It is the avoidable penalty from betting suboptimally even after M is fixed. That decomposition earns its place for the same reason the bias–variance decomposition does in supervised learning. It tells you which part of the error is structural and which part is fixable by better optimisation.

Corollary 3.15 (Representation search)**.** *Fix the evaluation distribution and let $\mathcal{M}$ be any nonempty family of admissible representations. Then*

$$\inf_{M\in\mathcal{M}} \inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i\, \delta_i(\pi) = \inf_{M\in\mathcal{M}} K_\rho(M).$$

If the infimum on the right is attained by $M^\star$, every cellwise optimal $M^\star$-based policy jointly minimises regret over $\mathcal{M}$. Conversely, every joint minimiser uses a representation attaining the minimum K_ρ and a cellwise optimal policy for that representation.

Proof. Apply Theorem 3.10 separately to every $M \in \mathcal{M}$, then take the infimum over $\mathcal{M}$. The statements about attained minima follow from Theorem 3.14. □

Interpretation. Regret minimisation can search over representations as well as policies. The family $\mathcal{M}$ must be declared. A family containing a representation that is injective on the histories in the evaluation support has minimum cost zero, so memory, state cardinality, architecture, latency, or another embodiment constraint must delimit a nontrivial search.

Example 3.16 (A tiny quotient task)**.** Suppose one mixed cell C_1 has $A_{C_1} = 0.30$ and $B_{C_1} = 0.20$, while a second cell C_2 is pure with $A_{C_2} = 0$ and $B_{C_2} = 0.25$. Then

$$K_\rho(M) = \min\{0.30, 0.20\} + \min\{0, 0.25\} = 0.20.$$

If an actual policy uses $q_{C_1} = 0.8$ and $q_{C_2} = 0$, then its regret is

$$0.30(1 - 0.8) + 0.20(0.8) + 0.25(0) = 0.22.$$

The extra 0.02 is

$$|0.30 - 0.20|\, |0.8 - 1| = 0.02.$$

The 0.20 is the irreducible cost of aliasing opposite bets inside C_1. The extra 0.02 is just the cost of choosing the wrong within-cell report, with nothing representational in it.

Corollary 3.17 (Unweighted forgetting on the informative region)**.** *In practice one may wish to ignore near-tie diagnostic tests whose margin m_i is close to zero, for instance when empirical estimates of p_i are noisy and tests near $\frac{1}{2}$ afford little signal. The following shows that restricting to sufficiently informative tests recovers an unweighted forgetting picture up to fixed constants.*

Fix $0 < \gamma \le \frac{1}{2}$ and let

$$\mathcal{X}_\gamma := \{i : m_i \ge \gamma\}.$$

Define the unweighted informative-region forgetting cost

$$K_\gamma(M) := \min\Big\{ \sum_{i\in B\cap\mathcal{X}_\gamma} \mu_i \;\Big|\; B \subseteq \{1, \ldots, n\},\ \text{and for every } C \in \mathcal{C}_M,\ y \text{ is constant on } (C \cap \mathcal{X}_\gamma) \setminus B \Big\}.$$

Then, with $c(\gamma) = \frac{4\gamma}{1+2\gamma}$*,*

$$c(\gamma)K_\gamma(M) \le \inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \mu_i \, \delta_i(\pi) \le K_\gamma(M).$$

Proof. On $\mathcal{X}_\gamma$, $\rho_i \in [c(\gamma), 1]$. Apply Theorem 3.10 on the restricted support $\mathcal{X}_\gamma$. The resulting weighted deletion cost lies between $c(\gamma)$ times the unweighted deletion mass and the unweighted deletion mass itself. □

Interpretation. This recovers the intuitive raw-forgetting picture. If you only look at tests that matter by at least γ, then weighted forgetting and ordinary forgetting differ only by fixed constants.

3.2 Multi-class generalisation

The binary case connects directly to Nayebi's betting setup, but the algebraic structure survives for any finite number of outcome classes.

Theorem 3.18 (Multi-class exact reduction)**.** *Assume a* K*-class setting,* $K \ge 2$*, in which each support point has an optimal label* $y_i \in \{1, \dots, K\}$*, an* M*-based policy assigns a class distribution* $q_C \in \Delta^{K-1}$ *to each cell* C*, and pointwise regret takes the weighted misreport form*

$$\delta_i(\pi) = \rho_i\bigl(1 - q_C^{(y_i)}\bigr) \quad \text{for } i \in C.$$

For $K = 2$ *this is Theorem 3.7 with* $q_C^{(1)} = q_C$*. For each cell* $C \in \mathcal{C}_M$ *and class* k*, define*

$$A_C^{(k)} := \sum_{i \in C,\, y_i = k} \mu_i \rho_i.$$

Then

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \, \delta_i(\pi) = \sum_{C \in \mathcal{C}_M} \Bigl(\sum_{k=1}^{K} A_C^{(k)} - \max_k A_C^{(k)} \Bigr).$$

The right-hand side is the minimum margin-weighted deletion cost needed to make y *single-valued on each cell, and it equals the cellwise weighted Bayes error.*

Proof. By the misreport form, the expected regret on cell C is

$$\sum_{k=1}^{K} A_C^{(k)}\bigl(1 - q_C^{(k)}\bigr) = \Bigl(\sum_k A_C^{(k)}\Bigr) - \sum_k A_C^{(k)} q_C^{(k)}.$$

To minimise this over $q_C \in \Delta^{K-1}$, one must maximise $\sum_k A_C^{(k)} q_C^{(k)}$, which is achieved by putting all mass on the class with the largest $A_C^{(k)}$. So the minimum per-cell regret is $\sum_k A_C^{(k)} - \max_k A_C^{(k)}$. To make y single-valued on C, one must delete every point not in the dominant class, costing $\sum_k A_C^{(k)} - \max_k A_C^{(k)}$. Summing over cells gives both sides. □

Interpretation. In each cell, the optimal policy picks the majority class, weighted by $\mu_i \rho_i$. The cost is the total weight of the non-majority points. The binary case is recovered at $K = 2$.

4 Freedom and the lifted bridge

This section introduces every Stack-Theoretic notion used in the bridge and its consequences in Section 5. No prior familiarity with Stack Theory is assumed. The representation-induced regret floor of Section 3 instantiates a general criterion for how well a learner generalises, namely *least commitment*. The vocabulary below expresses that claim. The electronic supplementary material states the imported results and proves them in full. A reader concerned only with the reinforcement-learning result can continue at Section 6.

4.1 Statements, extensions and freedom

A learner physically enacts *statements*, as its internal state is updated to reflect its learning. A statement that commits to little can be specialised in many ways. This is a reflection of both less resource use, and more general hypotheses. Freedom enumerates those compatible specialisations. Earlier papers call the same tally weakness. Absent information about what will be tested next, the freedom-maximising statement consistent with the observations leaves the largest set of future commitments open.

Now the formal version. Let Φ be a set of mutually exclusive *states*, and let a *program* be a set $p \subseteq \Phi$ of states (the situations in which p holds). A finite *vocabulary* $\mathfrak{v}$ is a finite set of programs.

Definition 4.1 (Embodied language)**.** The *embodied language* of $\mathfrak{v}$ is the set of satisfiable conjunctions of its programs,

$$L_{\mathfrak{v}} := \left\{ l \subseteq \mathfrak{v} \;:\; \textstyle\bigcap_{p\in l} p \neq \varnothing \right\}.$$

An element $l \in L_{\mathfrak{v}}$ is a *statement*. It holds in those states in $\bigcap_{p\in l} p$ (with the convention $\bigcap_{p\in\varnothing} p := \Phi$).

Definition 4.2 (Extension and freedom)**.** The *extension* of a statement $l \in L_{\mathfrak{v}}$ is the set of statements that contain it,

$$\mathrm{Ext}(l) := \{\, y \in L_{\mathfrak{v}} \;:\; l \subseteq y \,\}.$$

Its *freedom* is

$$w(l) := |\mathrm{Ext}(l)|.$$

Earlier work calls $w(l)$ the weakness of l. I retain w for continuity and write $l_1 <_w l_2$ when $w(l_1) < w(l_2)$. Then l_2 has more freedom than l_1.

A running example. Fix four states $\Phi = \{1, 2, 3, 4\}$ and a vocabulary of three programs $a = \{1, 2\}$, $b = \{2, 3\}$, $c = \{3, 4\}$. A subset of $\{a, b, c\}$ is a statement when its programs share a state, so

$$L_{\mathfrak{v}} = \{\, \varnothing,\ \{a\},\ \{b\},\ \{c\},\ \{a, b\},\ \{b, c\} \,\},$$

because $\{a, c\}$ and $\{a, b, c\}$ have empty intersection. Then

$$\mathrm{Ext}(\{a, b\}) = \{\{a, b\}\}, \qquad \mathrm{Ext}(\{a\}) = \{\{a\}, \{a, b\}\}, \qquad \mathrm{Ext}(\{b\}) = \{\{b\}, \{a, b\}, \{b, c\}\},$$

so $w(\{a, b\}) = 1 < w(\{a\}) = 2 < w(\{b\}) = 3$. Adding the program b to $\{a\}$ yields the more specific statement $\{a, b\}$, whose extension is *smaller*. The more a statement commits to, the fewer ways it can be completed. This order reversal is the whole content of Theorem 4.4 below.

4.2 Tasks and correct policies

A *task* fixes some inputs and the outputs that qualify as correct on them. A *policy* is itself a statement, and it is *correct* when, on the seen inputs, the completions it shares with them are the required outputs. Generalisation asks whether a policy correct on the seen inputs stays correct when new required outputs appear.

Definition 4.3 (Task and correct policy)**.** A $\mathfrak{v}$-*task* is a pair $\alpha = \langle I_\alpha, O_\alpha \rangle$ with inputs $I_\alpha \subseteq L_\mathfrak{v}$ and correct outputs $O_\alpha \subseteq \mathrm{Ext}(I_\alpha)$, where $\mathrm{Ext}(X) := \bigcup_{x \in X} \mathrm{Ext}(x)$. A policy is a statement $\pi \in L_\mathfrak{v}$, and it is *correct* for α when $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$. I write Π_α for the set of correct policies, and $\alpha \sqsubset \omega$ when ω extends α by requiring further outputs.

The bridge uses one specific task built from the finite evaluation support of Section 3. Each support point $x_i = (h_i, T_i)$ contributes one *unseen output* u_i, endowed with that point's label $y_i \in \{0, 1\}$ and weight $w_i := \mu_i \rho_i$. The collection $U = \{u_1, \ldots, u_n\}$ is the *unseen region* of the *lifted diagnostic task*, and a policy on it retains, in each representation-test cell, one label class. This happens to be the construction of Section 4.5 (Table 3). The warm-up in Theorem 3.9 and the worked example in Theorem 3.16 compute it on small instances. The running example above is the intuition to bear for "extension" and "freedom" throughout.

4.3 Freedom and generality

For a reader coming from machine learning, freedom has a familiar identity, and stating it removes most of the unfamiliarity of what follows.

Proposition 4.4 (Freedom measures generality)**.** *Fix a finite vocabulary* $\mathfrak{v}$*. For* $l, l' \in L_\mathfrak{v}$ *with* $l \subseteq l'$, $\mathrm{Ext}(l') \subseteq \mathrm{Ext}(l)$*. The map* $l \mapsto \mathrm{Ext}(l)$ *therefore reverses the commitment order, so a statement that commits to less admits every completion that any of its strengthenings admits, and possibly more. Consequently, for any task* α *(Theorem 4.3) the correct policies* Π_α *are those statements consistent with the seen input–output pairs (the* version space *of* α*), and every* $<_w$*-maximal member of* Π_α *is a member of that version space with maximal extension cardinality, an element of the G-boundary of classical candidate-elimination learning [33, 34].*

Proof. If $y \in \mathrm{Ext}(l')$ then $l' \subseteq y$, and $l \subseteq l' \subseteq y$ gives $l \subseteq y$, i.e. $y \in \mathrm{Ext}(l)$. Thus $\mathrm{Ext}(l') \subseteq \mathrm{Ext}(l)$ and $w(l') \leq w(l)$, so the map $l \mapsto \mathrm{Ext}(l)$ reverses the commitment order. The identification of Π_α with the version space is Theorem 4.3 read as a consistency condition. A correct policy reproduces every required output and admits no other on the seen inputs. "Maximally general" means maximal in the coverage order, where π is below π' when $\mathrm{Ext}(\pi) \subsetneq \mathrm{Ext}(\pi')$. If a $<_w$-maximal $\pi \in \Pi_\alpha$ had a strict coverage extension $\pi' \in \Pi_\alpha$, then $w(\pi) < w(\pi')$, contradicting $<_w$-maximality. So every $<_w$-maximal correct policy is coverage-maximal in the version space. □

Remark 4.5 (Cardinality, not inclusion). Freedom compares extension *sizes*, not extensions as sets. $w(l_1) > w(l_2)$ does not imply $\mathrm{Ext}(l_2) \subseteq \mathrm{Ext}(l_1)$, because two statements can be incomparable under inclusion while their extensions differ in cardinality. The order reversal in Theorem 4.4 holds along the commitment order $l \subseteq l'$, and the G-boundary can contain coverage-maximal policies that are not $<_w$-maximal. Freedom agrees with generality on direction and additionally ranks the boundary by how much each member leaves open. That extra resolution is what the freedom-optimality results in the electronic supplementary material exploit.

Table 2: Each Stack-Theoretic object used in the bridge with its closest standard machine-learning counterpart. The first block is exact (Theorem 4.4). Rows below the midrule are warnings, not correspondences.

Stack-Theoretic object	**Standard machine-learning counterpart**
Statement / policy π	Hypothesis h
Extension $\mathrm{Ext}(\pi)$	Coverage of h (what it admits)
Freedom $w(\pi) = \lvert\mathrm{Ext}(\pi)\rvert$	Generality of h (greater freedom = more general)
Correct policies Π_α	Version space (consistent hypotheses)
$<_w$-maximal correct policy	Maximally general consistent hypothesis in the G-boundary (least commitment)
Supplementary freedom-optimality result	Max-general consistent hypothesis is Bayes-optimal under a uniform prior over unseen positives
Weighted freedom deficit $Z - W^\star(M)$ on the lifted diagnostic task (Theorem 4.10)	Weighted Bayes error under the representation-test partition. Equality holds on the lift
Freedom vs. weighted Bayes error	*Not* the same axis off the lifted diagnostic task. Bayes error depends only on cell masses. Freedom depends on extension structure (Theorems 4.5 and 4.11)
Freedom vs. description length	*Not* the same axis (Theorem 4.6)

Interpretation. "Prefer the freedom-maximising correct policy" is the embodied-language form of the oldest bias in concept learning. Among hypotheses consistent with the data, choose a maximally general one. The freedom-optimality results in the electronic supplementary material make this optimal rather than merely traditional. Under a uniform prior over which unseen cases are tested, the maximally general consistent hypothesis is the one most likely to remain correct. Least commitment is Bayes-optimal generalisation under maximal ignorance.

Remark 4.6 (Freedom measures generality, not simplicity). A selection rule of the form "prefer the freedom-maximising consistent hypothesis" invites a reflex reading as Occam's razor or minimum description length. That reading is wrong here. Freedom measures *generality* (size of the extension), a different axis from *simplicity* (shortness of description), and the two disagree in general. The electronic supplementary material records that description-length minimisation can select a non-freedom-maximising policy in some vocabularies and is therefore not generalisation-optimal there. The criterion of this paper is the maximally-general-hypothesis criterion, not a Solomonoff-style description-length prior.

4.4 Weighted freedom and Enactive General Reinforcement Learning (EGRL)

When unseen outputs are not equally likely to be tested, the relevant quantity is *prior-weighted* freedom, in which each retained unseen output contributes its weight w_i rather than a unit (Theorem 4.7, with the underlying optimality proved in the electronic supplementary material). Under a uniform prior this reduces to ordinary freedom. Under the canonical prior $r_i = 1 - e^{-\mu_i \rho_i}$ the regret floor $K_\rho(M)$ acquires the exponential reading $e^{-K_\rho(M)}$ used in Theorem 5.4.

The setting in which these ideas were originally developed is Enactive General Reinforcement Learning (EGRL), the Stack-Theoretic translation of reward-labelled interaction into the task language. Interaction histories are split into positive and negative subhistories by a binary reward predicate, admissible policy pairs are ranked by a freedom-based pair proxy, and outputs that contradict otherwise good policies may be selectively deleted [8, 10, 11]. My theoretical contribution here is to the mathematical core of that programme, and the electronic supplementary material states the EGRL constructs it uses and the result that the freedom pair proxy is optimal. None of these constructs is assumed below beyond the definitions just given.

4.5 The lifted diagnostic task

Section 3 stated the identity in the language of fixed-distribution regret, following Nayebi [36]. I now restate that identity on one specific Stack-Theoretic object, the lifted diagnostic task introduced in Section 4. The scope is narrower than a general freedom-Bayes-error equivalence. The lift puts one unseen output over each evaluation point and restricts admissible policies to independent per-cell binary retain/delete choices. That restriction removes the cross-cell extension structure that normally gives freedom more resolution than class-mass arithmetic. In this restricted task, the representation-induced regret floor $K_\rho(M)$ becomes the deficit $Z - W^\star(M)$ between total weight and maximum retained weighted freedom. This is why Section 5.2 can read it as a generalisation probability rather than as an analogy. It also connects an RL diagnostic to the broader programme that links generalisation-optimal learning to selective memory, causal structure, and biological organisation. Concretely, the finite diagnostic support becomes the unseen region U of the lifted task, the margin weights $w_i = \mu_i \rho_i$ become the weights of a prior-weighted freedom score on that lifted support, and the minimum-regret policy becomes the freedom-maximising admissible policy under the lift's admissibility restriction. Section 4 introduced these notions. Theorem 4.10 states the scoped correspondence.

4.5.1 Notation and prerequisites

This subsection gives self-contained definitions sufficient to verify the Stack-Theoretic form of the bridge (Theorem 4.10). The reader familiar with Stack Theory may skip ahead.

A *lifted diagnostic task* is defined from the finite evaluation support as follows. Construct a set $U = \{u_1, \ldots, u_n\}$ of *unseen outputs*, one for each support point $x_i = (h_i, T_i)$. Each unseen output u_i inherits the label $y_i \in \{0, 1\}$ and weight $w_i := \mu_i \rho_i$ from its corresponding support point. A *policy on the lifted task* selects, for each representation-test cell $C \in \mathcal{C}_M$, which label class to retain. Its *extension*

$$\mathrm{Ext}(\vartheta) \cap U := \{u_i \in U : y_i \text{ equals the label class retained in cell } C \ni i\}$$

is the set of unseen outputs whose label matches the policy's selection. The policy is *admissible* if it retains outputs from at most one label class per cell. Because the lift treats these retain/delete choices as independent across cells, the feasible retained sets have a per-cell product form. A general embodied language need not have that product form, and cross-cell extension constraints can block the cellwise Bayes-optimal selection.

This vocabulary is standard in the Stack-Theoretic framework [6, 11] but is used here only in the restricted form above. Table 3 gives the correspondence.

Table 3: Notation mapping between the Stack-Theoretic and RL sides.

Stack-Theoretic side	**RL side**
Truth set of a diagnostic statement	Representation-test cell C
Child policy ϑ	Policy π
Label class selection per cell	Report probability q_C
Weight not retained, $w_i \mathbf{1}\{i \notin \mathrm{Ext}(\vartheta)\}$	Normalised regret δ_i
Freedom deficit $Z - W^\star(M)$	Min regret $K_\rho(M)$
Freedom-maximising admissible policy	Optimal policy per cell

4.5.2 Weighted freedom and the bridge

Definition 4.7 (Prior-weighted freedom)**.** Let $U = \{u_1, \dots, u_n\}$ be a finite unseen region with positive weights $w_1, \dots, w_n$. For any policy ϑ, let $\mathrm{Ext}(\vartheta)$ denote its extension in the embodied language (Theorem 4.2), i.e. the set of all outputs it is compatible with [6, 11]. Define its prior-weighted freedom on U by

$$W(\vartheta) := \sum_{u_i \in \mathrm{Ext}(\vartheta) \cap U} w_i.$$

Ordinary freedom tallies unseen continuations left open by a policy. Weighted freedom sums their weights. This resembles the information bottleneck because low-weight exclusions cost less [52].

Remark 4.8 (This is not a new Stack-Theoretic primitive)*.* The thesis [11] defines the more general quantity

$$w_P(\pi) = P\big(S \subseteq \mathrm{Ext}(\pi) \cap U\big)$$

for an arbitrary prior P over subsets $S \subseteq U$. The score $W(\vartheta)$ above is just the finite product-prior specialisation that becomes convenient after taking logs [11]. So this section is a translation of existing Stack-Theoretic machinery, not a replacement for it.

Proposition 4.9 (Independent nonuniform priors give weighted freedom)**.** *Assume each unseen output $u_i \in U$ becomes relevant independently with probability $0 \le r_i < 1$. Then, for any correct child policy ϑ,*

$$\log \Pr(\vartheta \text{ generalises}) = \sum_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} \log(1 - r_i).$$

Equivalently, maximising generalisation probability is the same as maximising

$$\sum_{u_i \in \mathrm{Ext}(\vartheta) \cap U} \big(-\log(1 - r_i)\big).$$

So the Bayes-optimal rule is weighted freedom maximisation with weights

$$w_i := -\log(1 - r_i).$$

Proof. Generalisation occurs when every unseen output omitted by ϑ fails to become relevant. Independence gives the product

$$\Pr(\vartheta \text{ generalises}) = \prod_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} (1 - r_i).$$

Taking logs yields the display. The remaining term depends on ϑ only through the retained weighted support. $\square$

This is the weighted version of Bennett's earlier sufficiency theorem. Uniform priors recover ordinary freedom. Biased priors recover weighted freedom. The uniform case is the no-free-lunch baseline. If you know nothing about which unseen outputs matter, you keep as many open as possible [53].

Corollary 4.10 (Exact Stack-Theoretic form of the bridge)**.** *Let $w_i := \mu_i \rho_i$. Build a lifted diagnostic task whose unseen outputs are $u_1, \dots, u_n$, one for each support point x_i. Restrict admissible policies so that, on each representation-test cell $C \in \mathcal{C}_M$, the policy retains outputs from at most one label class. Let*

$$Z := \sum_{i=1}^{n} w_i, \qquad W^\star(M) := \max_{\vartheta \text{ admissible}} W(\vartheta).$$

Then

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \, \delta_i(\pi) = Z - W^\star(M).$$

Moreover, choosing independent prior probabilities

$$r_i := 1 - e^{-w_i}$$

turns $W^\star(M)$ into the Bayes-optimal weighted freedom score of the lifted task.

Proof. On the lifted task, retaining a support point u_i means not forgetting the corresponding diagnostic point x_i. Admissibility means each cell keeps at most one label class, and the lifted task imposes no constraint between cells. So maximising retained weighted mass is the same optimisation problem as minimising deleted weighted mass. Hence

$$W^\star(M) = Z - K_\rho(M).$$

Now apply Theorem 3.10. The final statement follows from Theorem 4.9 and the choice $r_i = 1 - e^{-w_i}$. $\square$

Proposition 4.11 (Freedom deficit dominates partition Bayes error)**.** *Let A_C and B_C denote the weighted masses of the two label classes in cell C, and let $Z = \sum_C (A_C + B_C)$. Embed the labelled support of Theorem 4.10 in a Stack-Theoretic task whose admissible policies $\mathcal{A}_M$ form a nonempty finite family with retained sets lying within the per-cell product selections, meaning that each admissible policy retains outputs from at most one label class in each cell. Define*

$$W^\star_{\mathcal{A}}(M) := \max_{\vartheta \in \mathcal{A}_M} \sum_{u_i \in \mathrm{Ext}(\vartheta) \cap U} w_i.$$

Then

$$W^\star_{\mathcal{A}}(M) \le \sum_C \max\{A_C, B_C\}, \qquad Z - W^\star_{\mathcal{A}}(M) \ge \sum_C \min\{A_C, B_C\} = K_\rho(M),$$

and the two hold with equality when some admissible policy retains weighted mass $\sum_C \max\{A_C, B_C\}$. The lifted diagnostic task of Theorem 4.10 meets that condition, so equality holds there. Strict inequality holds whenever cross-cell extension constraints rule out every cellwise Bayes-optimal retained-label selection.

Proof. Fix $\vartheta \in \mathcal{A}_M$ and write $R_\vartheta := \mathrm{Ext}(\vartheta) \cap U$. By hypothesis R_ϑ meets each cell C in outputs bearing at most one label, so the weight it draws from C is A_C, B_C, or zero, and in every case at most $\max\{A_C, B_C\}$. Summing over cells gives

$$\sum_{u_i \in R_\vartheta} w_i \le \sum_C \max\{A_C, B_C\},$$

and taking the maximum over $\mathcal{A}_M$ gives the first display. Subtracting from $Z = \sum_C (A_C + B_C)$ and using the identity $A_C + B_C - \max\{A_C, B_C\} = \min\{A_C, B_C\}$ gives the second, whose right-hand side is $K_\rho(M)$ by Theorem 3.10. The two displays differ by the constant Z, so they attain equality together, and each does so when the bound $\sum_C \max\{A_C, B_C\}$ is realised by some admissible policy. On the lift every per-cell product selection is admissible, so the policy retaining the heavier label class in each cell is available and realises the bound, which recovers Theorem 4.10. If instead no admissible policy retains the heavier label class in every cell, the bound is not realised and both displays are strict. □

Interpretation. Bayes error prices each cell on its own, so it can always take the cheaper option in every cell at once. Freedom cannot, because the embodied language decides which combinations of per-cell choices are expressible. When the language couples cells, the best expressible policy falls short of the cellwise optimum and the deficit exceeds the Bayes error by that gap. Two tasks can therefore possess identical cell masses, identical $K_\rho(M)$, and different freedom deficits. Theorem 4.12 exhibits this on a four-point support.

Example 4.12 (A two-cell witness for strict inequality)**.** Take four support points x_1, x_2, x_3, x_4 under a single test, with masses $\mu = (0.3, 0.2, 0.1, 0.4)$ and deterministic conditionals $p_i \in \{0, 1\}$, so every margin weight is $\rho_i = 1$ and $w_i = \mu_i$. The optimal labels are $y = (1, 0, 1, 0)$. The representation assigns h_1, h_2 to one state and h_3, h_4 to another, giving two cells $C_1 = \{x_1, x_2\}$ and $C_2 = \{x_3, x_4\}$ with masses

$$A_{C_1} = 0.3, \quad B_{C_1} = 0.2, \quad A_{C_2} = 0.1, \quad B_{C_2} = 0.4,$$

so $Z = 1$ and $K_\rho(M) = 0.2 + 0.1 = 0.3$. On the lift, the optimal admissible policy retains label 1 in C_1 and label 0 in C_2, so $W^\star_{\mathrm{lift}}(M) = 0.3 + 0.4 = 0.7$ and the deficit is $Z - W^\star_{\mathrm{lift}}(M) = 0.3 = K_\rho(M)$.

Now embed the same labelled support in an embodied language that couples the cells. Fix $\Phi = \{1, 2, 3, 4\}$ and the vocabulary $\mathfrak{v} = \{c_1, c_2, g_1, g_0\}$ with

$$c_1 = \{1, 2\}, \quad c_2 = \{3, 4\}, \quad g_1 = \{1, 3\}, \quad g_0 = \{2, 4\}.$$

The program c_j tags cell C_j and the program g_b tags label class b. The unseen outputs are the statements

$$u_1 = \{c_1, g_1\}, \quad u_2 = \{c_1, g_0\}, \quad u_3 = \{c_2, g_1\}, \quad u_4 = \{c_2, g_0\},$$

each inheriting the label and weight of its support point. Since $c_1 \cap c_2 = \varnothing$ and $g_1 \cap g_0 = \varnothing$, the language $L_\mathfrak{v}$ has nine statements, namely $\varnothing$, the four singletons, and the four outputs. A policy ϑ retains u_i when $\vartheta \subseteq u_i$, and $\mathcal{A}_M$ is the set of statements retaining at most one label class per cell, as in Theorem 4.10. Enumerating $L_\mathfrak{v}$ gives six admissible policies. Each output retains itself alone, $\{g_1\}$ retains $\{u_1, u_3\}$ with weight 0.4, and $\{g_0\}$ retains $\{u_2, u_4\}$ with weight 0.6. The other three statements are inadmissible, since $\{c_1\}$ retains both of u_1, u_2, $\{c_2\}$ retains both of u_3, u_4, and $\varnothing$ retains all four. Every admissible retained set is a per-cell product selection, so this family satisfies the hypothesis of Theorem 4.11. The cellwise Bayes-optimal selection $\{u_1, u_4\}$ needs $\vartheta \subseteq u_1 \cap u_4 = \varnothing$, and $\varnothing$ is inadmissible, so no admissible policy attains it. Hence

$$W^\star_{\mathcal{A}}(M) = 0.6, \qquad Z - W^\star_{\mathcal{A}}(M) = 0.4 > 0.3 = K_\rho(M).$$

The two embeddings have identical cell masses and the same $K_\rho(M)$, and their freedom deficits differ by 0.1.

Interpretation. Bayes error prices each cell on its own, so it bets label 1 in C_1 and label 0 in C_2. In this language the only programs shared across cells are g_1 and g_0, so committing to a label in one cell commits to it in the other. Freedom depends on that coupling, and cell masses do not. That is the resolution the deficit gains off the lift.

Interpretation. Regret is the weighted unseen future that the representation forced you to discard. The earlier sufficiency theorem holds that generalisation is about how much unseen future you leave open. The bridge holds that regret is the part of that weighted future lost to aliasing.

Remark 4.13 (Weighted freedom is not generalisation-optimal in the unconditional sense). The Sufficiency and Necessity theorems for unweighted freedom [11, 15] establish optimal generalisation under maximal ignorance. The freedom-maximising correct policy maximises generalisation probability without an added weighting.

Theorem 4.9 is a different beast. It asserts that if you adopt independent relevance probabilities r_i for the unseen outputs, then the rule that maximises generalisation probability under that adopted prior is weighted freedom maximisation with weights $w_i = -\log(1 - r_i)$. The optimality is conditional on the prior. Adopting any non-uniform prior is itself a commitment about which unseen futures matter, so weighted freedom can specialise the agent to that prior at the cost of generalisation under any other.

Two transformations must be separated. Freedom is invariant under function-preserving parameter reparameterisations once the embodied language and its measure are fixed [12]. Changing the embodied vocabulary changes which refinements qualify as completions, so the numerical value $|\mathrm{Ext}(\pi)|$ can change. The optimality rule remains the same inside every finite vocabulary. The Sufficiency proof gives

$$\Pr(\pi \in \Pi_\omega \mid \alpha) = \frac{2^{|\mathrm{Ext}(\pi)\cap U|}}{2^{|U|}},$$

which is monotone in $|\mathrm{Ext}(\pi) \cap U|$. Hence the freedom-maximising correct policy is generalisation-optimal under maximal ignorance in each embodied language.

Alternative rules do not have that property. The simplicity rule "pick the shortest correct policy" is provably non-optimal in some vocabularies because the shortest correct policy can fail to be freedom-maximising there [11, 15, Propositions on simplicity sub-optimality and subjectivity of description length]. Weighted freedom "maximise $\sum_{u_i \in \mathrm{Ext}(\vartheta)\cap U} w_i$" may be optimal under a specific prior P, but the prior is pinned to the unseen outputs u_i of a particular language. Re-encode the agent and the unseen region changes, the weights have to be re-specified for the new u_i, and whether the new weighted rule is optimal in the new language depends on whether the new prior matches the new environment. Plain freedom's optimality survives encoding changes. Alternatives' effectiveness does not.

Here I use the weighted form because an exact bridge to RL demands it. They use weights, so must I. The RL side conveys the weights ρ_i through normalised regret. Weighted freedom records that commitment. I do not promote it as a better generalisation criterion than unweighted freedom because its optimality depends on the chosen prior.

The EGRL and neural completion comparison is in the electronic supplementary material.

5 Consequences of the bridge

The identity is a proof technique, not just a dictionary. This section derives results via the bridge to illustrate its utility. I separate results that follow from the reduction alone (Section 5.1) from those that additionally import prior Stack-Theoretic machinery (Section 5.2).

5.1 Direct corollaries of the reduction

The following results use only Theorem 3.10 and basic partition combinatorics. No Stack-Theoretic imports are required.

5.1.1 A regret-based preorder on abstractions

The bridge induces a natural ordering on representations.

Proposition 5.1 (Regret preorder)**.** *Define $M_1 \preceq_\rho M_2$ iff $K_\rho(M_1) \geq K_\rho(M_2)$. This relation is a preorder on representations and a total order after quotienting representations by equal K_ρ value. It is consistent with the partition refinement lattice. If $\mathcal{C}_{M_2}|_{I_+}$ refines $\mathcal{C}_{M_1}|_{I_+}$, then $K_\rho(M_2) \leq K_\rho(M_1)$. The inequality is strict whenever the refinement splits at least one mixed cell on I_+ into subcells in which the minority class flips (i.e. the label achieving $\min\{A, B\}$ differs between subcells).*

Proof. If $\mathcal{C}_{M_2}$ refines $\mathcal{C}_{M_1}$, then every cell $C_1 \in \mathcal{C}_{M_1}$ is partitioned into subcells $C_2^{(j)} \in \mathcal{C}_{M_2}$. By the cellwise formula,

$$\min\{A_{C_1}, B_{C_1}\} \geq \sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\},$$

since $\min\{a + a', b + b'\} \geq \min\{a, b\} + \min\{a', b'\}$ for non-negative reals. Summing gives $K_\rho(M_1) \geq K_\rho(M_2)$. For strictness, suppose a mixed cell C_1 with $A_{C_1} > B_{C_1}$ splits into subcells C_2, C_2' where $A_{C_2} > B_{C_2}$ but $A_{C_2'} < B_{C_2'}$. Then

$$\min\{A_{C_1}, B_{C_1}\} = B_{C_2} + B_{C_2'} > B_{C_2} + A_{C_2'} = \min\{A_{C_2}, B_{C_2}\} + \min\{A_{C_2'}, B_{C_2'}\},$$

so the contribution from this cell is strictly smaller after refinement. □

Interpretation. This gives the state-abstraction and bisimulation communities a new tool, a regret-based preorder on abstractions that complements the existing bisimulation-metric order of Ferns, Panangaden, and Precup [19]. Finer representations have lower or equal K_ρ, and the ordering indicates when refinement helps. The ordering also complements the Richens–Everitt qualitative characterisation [43]. Their result affirms that robust agents must learn causal models. The regret ordering quantifies how much it costs if the learned representation falls short of the causal partition.

5.1.2 Diagnostic quotient convergence in RL language

The regret ordering has an endpoint at the quotient level. At zero K_ρ, every positive-weight cell is pure, and the coarsest pure diagnostic quotient is unique. For history representations, there is one extra feasibility constraint: the same history code must be reused across tests. This gives a diagnostic quotient convergence theorem stated entirely in RL language.

Theorem 5.2 (Diagnostic quotient convergence)**.** *Fix a finite evaluation support $(\mathcal{X}, \mu, y, \rho)$ and let $I_+ = \{i : \mu_i \rho_i > 0\}$ be the informative support. Let $\mathcal{T}_+ := \{T_i : i \in I_+\}$. Define $\mathcal{P}_+^\star := \big\{\{i \in I_+ :$*

$T_i = t,\, y_i = b\} : t \in \mathcal{T}_+,\, b \in \{0,1\}\big\} \setminus \{\varnothing\}$, *the coarsest partition of* I_+ *that separates test values and optimal labels. For any partition* $\mathcal{Q}$ *of* I_+ *whose cells refine test value, define*

$$K_\rho(\mathcal{Q}) := \sum_{C\in\mathcal{Q}} \min\Big\{ \sum_{i\in C, y_i=1} \mu_i\rho_i,\ \sum_{i\in C, y_i=0} \mu_i\rho_i \Big\}.$$

1. $K_\rho(\mathcal{Q}) = 0$ *if and only if* $\mathcal{Q}$ *refines* $\mathcal{P}_+^\star$.
2. $\mathcal{P}_+^\star$ *is the unique coarsest zero-cost diagnostic quotient on* I_+.
3. *For an* M*-based history representation,* $K_\rho(M) = 0$ *if and only if* $\mathcal{C}_M|_{I_+}$ *refines* $\mathcal{P}_+^\star$. *If* $\mathcal{P}_+^\star$ *is induced by some history representation, then every minimal zero-cost* M *induces* $\mathcal{P}_+^\star$ *up to relabelling. Without that realisability condition, minimal zero-cost history representations induce minimal feasible refinements of* $\mathcal{P}_+^\star$*, and need not be unique.*

Proof. For a partition cell C, the cell contribution to $K_\rho(\mathcal{Q})$ is $\min\{A_C, B_C\}$ with A_C and B_C the positive-weight masses of the two labels. This contribution is zero when all positive-weight points in C have the same optimal label. Since the cells of $\mathcal{Q}$ refine test value, this is equivalent to every cell of $\mathcal{Q}$ being contained in a cell of $\mathcal{P}_+^\star$. That proves part 1. For part 2, $\mathcal{P}_+^\star$ is itself zero-cost because each of its cells is pure, and part 1 confirms that every zero-cost diagnostic quotient refines it. For part 3, apply parts 1 and 2 to the induced quotient $\mathcal{Q} = \mathcal{C}_M|_{I_+}$. The last sentence follows because M maps histories globally, while $\mathcal{P}_+^\star$ is a quotient of history-test support points. When the coarsest diagnostic quotient cannot be realised by any single history code, the minimal zero-cost representation must be a feasible refinement instead. □

Interpretation. The unique endpoint exists at the diagnostic quotient level. A history representation can be forced to refine that endpoint when the same history participates in several tests with different labels. The proof uses only the cellwise decomposition from Theorem 3.10 and basic partition logic.

5.1.3 Rate-distortion interpretation

$K_\rho(M)$ has a natural information-theoretic reading. The representation M is a lossy compression of histories. The margin-weighted forgetting cost $K_\rho(M)$ is the residual relevant information that the bottleneck has destroyed, weighted by diagnostic importance. In the language of the information bottleneck [52], $K_\rho(M)$ is the Bayes error of the sufficient statistic induced by M, expressed as a margin-weighted deletion cost. The bridge theorem identifies this as the regret cost of that compression. Representations with lower K_ρ are better compressions in the decision-relevant sense. They preserve distinctions that affect regret and discard distinctions that do not.

5.1.4 Stability under margin estimation error

In practice the conditional probabilities p_i are estimated from data, so the margin weights ρ_i are known only approximately. The following shows that K_ρ is Lipschitz in the margin estimates, so small estimation errors produce small diagnostic errors.

Proposition 5.3 (Lipschitz stability of K_ρ)**.** *Let* $\hat\rho_1, \ldots, \hat\rho_n$ *be perturbed margin weights. Define* $\hat K_\rho(M)$ *by replacing* ρ_i *with* $\hat\rho_i$ *in Theorem 3.8. Then*

$$\big|K_{\hat\rho}(M) - K_\rho(M)\big| \ \le\ \sum_{i=1}^{n} \mu_i\, |\hat\rho_i - \rho_i|.$$

Moreover, ρ_i is a Lipschitz function of p_i with constant 4 on $(0,1)$ (the one-sided derivatives satisfy $|d\rho/dp| \le 4$ on $(0,\frac{1}{2}) \cup (\frac{1}{2},1)$ and the two pieces agree at the kink), so estimation error in the conditional probabilities propagates as

$$\left|K_{\hat{\rho}}(M) - K_{\rho}(M)\right| \;\le\; 4\sum_{i=1}^{n} \mu_i\,|\hat{p}_i - p_i|.$$

Proof. For each cell C, define

$$\hat{A}_C := \sum_{i \in C,\, y_i = 1} \mu_i \hat{\rho}_i, \qquad \hat{B}_C := \sum_{i \in C,\, y_i = 0} \mu_i \hat{\rho}_i.$$

For non-negative real numbers,

$$|\min\{a', b'\} - \min\{a, b\}| \le |a' - a| + |b' - b|.$$

Therefore

$$\left|\min\{\hat{A}_C, \hat{B}_C\} - \min\{A_C, B_C\}\right| \le |\hat{A}_C - A_C| + |\hat{B}_C - B_C| \le \sum_{i \in C} \mu_i |\hat{\rho}_i - \rho_i|.$$

Summing over cells gives the first bound. For the second, note that for $p > \frac{1}{2}$, $\rho = (2p-1)/p$ and $d\rho/dp = 1/p^2 \le 4$. For $p < \frac{1}{2}$, $\rho = (1-2p)/(1-p)$ and $|d\rho/dp| = 1/(1-p)^2 \le 4$. The two pieces agree at $p = \frac{1}{2}$, where $\rho = 0$, so ρ is Lipschitz with constant 4 on all of $(0,1)$. Hence $|\hat{\rho}_i - \rho_i| \le 4|\hat{p}_i - p_i|$. □

Interpretation. The diagnostic $K_\rho(M)$ degrades gracefully under estimation noise. With N samples per support point, standard concentration gives $|\hat{p}_i - p_i| = O(N^{-1/2})$, so $|K_{\hat{\rho}} - K_\rho| = O(N^{-1/2})$ as well.

5.2 Cross-framework consequences

The generalisation result below combines Theorem 3.10 with prior Stack-Theoretic machinery. The expected-free-energy paragraph records an external limit on that comparison.

5.2.1 Expected free energy

The regret identity does not induce a general expected-free-energy ordering. External work shows that expected free energy can agree with freedom under a declared uniform-viability construction in which the embodied outcome space, forecast, and preferences are fixed. Uniform preferences alone do not preserve the order because unequal forecasts can reverse it, and varying preferences can induce any strict order [14]. I therefore do not derive an expected-free-energy corollary from K_ρ here. Any such relation needs assumptions external to the regret identity and to the freedom bridge.

5.2.2 Generalisation probability from regret

The following corollary converts the regret floor into a generalisation probability under the canonical independent demand prior, giving K_ρ a predictive semantics that the RL identity alone does not provide.

Corollary 5.4 (Generalisation probability from regret)**.** *In the setting of Theorem 4.10, choose independent relevance probabilities $r_i = 1 - e^{-\mu_i \rho_i}$ as in Theorem 4.10. Under this prior, the probability that the optimal M-based policy generalises to a new set of diagnostic demands is*

$$\Pr(\textit{optimal policy generalises}) = e^{-K_\rho(M)}.$$

Proof. By Theorem 4.9, generalisation occurs when no deleted unseen output becomes relevant. The optimal admissible policy deletes the cheaper label class in each cell, removing total weight $K_\rho(M)$ from the unseen region. Under the independent prior with $r_i = 1 - e^{-w_i}$, the generalisation probability of any admissible policy ϑ is

$$\prod_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} (1 - r_i) = \prod_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} e^{-w_i} = e^{-\sum_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} w_i}.$$

For the optimal policy, the exponent is $K_\rho(M)$. □

Interpretation. This is the payoff of the bridge as a bidirectional connection, not merely a dictionary. The RL side treats $K_\rho(M)$ as a regret floor. The Stack-Theoretic side, via the independent-prior generalisation model, converts that regret floor into a generalisation probability. Neither framework alone gives both quantities. A representation with $K_\rho = 0.02$ generalises with probability $e^{-0.02} \approx 0.98$ under the canonical prior. One with $K_\rho = 0.20$ generalises with probability $e^{-0.20} \approx 0.82$. The exponential sensitivity to K_ρ gives the diagnostic empirical force. Small differences in irreducible regret correspond to measurable differences in generalisation reliability.

6 Recurrent Proximal Policy Optimization (PPO) representation search

These results have an immediate practical application, in facilitating for the optimisation of representation for the minimisation of regret. Theorem 3.15 states that, within any declared family, selecting a representation with minimum K_ρ also selects the smallest attainable normalised regret. To falsify this claim I test it on learned recurrent states under a delay shift. The question is whether K_ρ can choose among source-trained memories before target rewards or target labels are available. I use two complementary designs. A *fixed world* is evaluated without conditioning on selector disagreement, so selector agreements remain. A *disagreement world* is retained only when selectors choose different candidates under pre-target screening and both choices remain unchanged when portions of the unlabelled target histories are omitted. Fixed worlds estimate candidate-ranking quality and average deployment benefit. Disagreement worlds test which selector chooses better when their recommendations differ robustly before hidden target outcomes are revealed.

6.1 Delayed-cue POMDP and candidate family

Each episode has one hidden class in $\{1, 2, 3, 4\}$ and three later binary diagnostic tests. Each test maps two classes to answer 0 and two to answer 1, while the three-answer vector distinguishes all four classes. Rare and common episodes are two visible *regimes* with different cue reliabilities and placements. At a test, *decision margin* is the normalised value lost by choosing the worse binary answer. Rare-regime episodes contain three reliable cues near the start of the episode. Common-regime episodes contain three weaker cues immediately before the first test. The rare regime therefore has a high decision margin but a long memory demand, while the common regime has a lower decision margin and a short memory demand. Source delays are 3, 5, and 7 steps. Target delays are 12, 20, and 28 steps. Cue reliabilities and test mappings stay fixed between source and target, so only memory delay shifts. Answers earn reward at the three tests but do not alter later observations. This isolates memory and representational aliasing from exploration and transition control.

For each world I train six recurrent PPO agents on source episodes [46]. The recurrent encoder is either a gated recurrent unit (GRU) or long short-term memory (LSTM) network with 32 hidden units. Training episodes are sampled from the natural source distribution, a rare-regime-enriched distribution, or a common-regime-enriched distribution. Snapshots at updates 60 and 120 give 12 encoders. One hidden state immediately before the first test is compressed and reused for all three tests. Each encoder is combined with five source-fitted *quantisers*, each mapping its hidden state to one of six categorical states divided between regimes as $5+1$, $4+2$, $3+3$, $2+4$, or $1+5$. This gives 60 candidate representations per world.

Each encoder uses one deterministic *affine row frame*, an affine coordinate system fitted from source hidden rows and reused across all five allocations. The resulting codes are unchanged by dense invertible affine recodings or shifts of hidden coordinates. This removes dependence on an arbitrary orientation or scale of recurrent-state coordinates. *Source validation* is candidate admission using an independent source episode set. Candidates are admitted when reward on that set lies within 0.025 of the best candidate in that world, with at least 12 candidates admitted. This holds source competence approximately fixed before the representation criterion is applied.

The fixed design contains two independent replications of 24 worlds, giving 48 worlds and 2,880 candidates. The disagreement design screens 224 further worlds and 13,440 candidates to obtain two independent 24-world cohorts where selector choices differ and remain unchanged across eight target-history folds. Full environment, optimiser, episode-set, geometry, admission, screening, and audit settings are in the electronic supplementary material.

I use *target truth* for hidden target classes, posterior class probabilities, correct answers, rewards, and regret.

6.2 Selection before target truth is revealed

Source calibration uses independent labelled source episodes to estimate cue reliability in each regime. A *target panel* is an input-only set of target episodes used for selection. It is generated and stored without hidden classes, correct answers, posterior probabilities, rewards, or regret. For target history i and binary test j, source calibration gives estimated answer values $\widehat{V}_{ij}(b)$ for $b \in \{0,1\}$. Define

$$\widehat{y}_{ij} = \arg\max_{b\in\{0,1\}} \widehat{V}_{ij}(b), \qquad \widehat{\rho}_{ij} = 1 - \frac{\min_b \widehat{V}_{ij}(b)}{\max_b \widehat{V}_{ij}(b)}. \tag{1}$$

For candidate representation M, the source-calibrated target-input estimate is

$$\widehat{K}_{\rho,T}(M) = \frac{1}{3N}\sum_{j=1}^{3}\sum_{c}\min_{b\in\{0,1\}}\sum_{i:M(h_i)=c}\widehat{\rho}_{ij}\,\mathbf{1}\{\widehat{y}_{ij}\neq b\}. \tag{2}$$

I use KBT as a label[4], not an acronym, for this target-input selector. A *cell policy* assigns one binary answer per representation-test cell. KBT chooses each cell's estimated weighted-conflict minimiser.

[4] It arose because I had the variable K, used in experiment candidate part B, and this was the transductive variant.

Raw impurity receives the same target histories and the same source-estimated optimal answers, but replaces every margin weight with one. After raw impurity selects a representation, the primary regret comparison fits the same margin-weighted cell policy used after KBT selection. Any regret difference therefore comes from representation choice rather than a different policy rule. I also compare source K_ρ, source validation, uniform random selection over admitted candidates, and a *target oracle* that minimises fresh target K_ρ after target truth is opened.

Fixed worlds retain all selector agreements. Their primary endpoint is the paired within-world difference between the Spearman correlation of each selector score with fresh target K_ρ across admitted candidates. *Rank-one deployment* deploys a selector's first-ranked candidate. Its fresh target K_ρ is the deployment endpoint. Disagreement screening uses only source data and the input-only target panel. A *leave-one-panel-fold-out fit* omits one of eight disjoint target-panel folds. A choice is *fold-stable* when unchanged across all eight fits. A world is retained when pooled KBT and raw impurity select different fold-stable candidates. Sixteen source-calibration folds are recorded as an additional stability audit and do not alter screening or selection.

A *source stage* is all work completed without target truth. It is *locked* when candidate identities, cell policies, panel and quantiser hashes, fold decisions, and selector hashes are written. These records are the *lock*. A *fresh target bank* is disjoint from the target panel and evaluated only after every source stage is locked. Nonselected screening worlds receive no target evaluation. The audit independently recomputes candidate scores, selectors, policies, screening decisions, geometry, and chronology.

6.3 Results

Across the 48 fixed worlds, mean within-world Spearman correlation with fresh target K_ρ was 0.959470 for KBT and 0.910559 for raw impurity. The paired gain was 0.048911, with a 95% bootstrap interval of $[0.032299, 0.066819]$. KBT ranked candidates better in 44 worlds and raw impurity in four, with one-sided sign-flip $p < 10^{-6}$. For rank-one deployment, KBT selected mean target K_ρ 0.100015 against 0.101843 for raw impurity. The reduction was 0.001828, with interval $[0.000748, 0.003016]$ and $p = 0.000720$. The selectors chose the same representation in 25 worlds. KBT won 20 of the remaining 23 and lost three. Target regret fell by 0.003098, with interval $[0.001795, 0.004601]$, and target success rose by 0.005949, with interval $[0.003872, 0.008163]$.

Disagreement replication A retained 24 worlds from 128 screened worlds. Replication B retained 24 from 96. Across the resulting 48 worlds, KBT selected mean target K_ρ 0.090744 against 0.096840 for raw impurity. The reduction was 0.006096, or 6.3%, with interval $[0.004713, 0.007485]$. KBT won 43 worlds and lost five. Replication A separately reduced target K_ρ by 0.006581, with interval $[0.004870, 0.008294]$ and exact $p = 4.77 \times 10^{-7}$. Replication B reduced it by 0.005610, with interval $[0.003498, 0.007809]$ and exact $p = 1.91 \times 10^{-5}$. Their Fisher-combined value was $p = 2.40 \times 10^{-10}$.

Target normalised regret in disagreement worlds fell from 0.099838 under raw-impurity selection to 0.091498 under KBT. The reduction was 0.008340, or 8.4%, with interval $[0.006753, 0.009916]$. Sampled target success rose from 0.528626 to 0.541670. The gain was 0.013044, or 1.30 percentage points, with interval $[0.010697, 0.015593]$. KBT increased success in 47 worlds and tied in one. The native recurrent PPO actors selected by KBT also increased target return by 0.012767, with interval $[0.0067, 0.0193]$. Its one-sided sign-flip value was approximately 8×10^{-5}.

Across all 2,880 candidates in the fixed worlds, estimated and fresh target K_ρ had mean within-world Spearman correlation 0.962606. Mean absolute error was 0.006127, median error 0.004987, and maximum error 0.031374. KBT selected the target-oracle representation in 39 of 48 fixed worlds, with mean target-oracle K_ρ gap 0.000589. The median candidate occupied all 18 possible representation-test cells, and 2,686 of 2,880 candidates occupied all 18, so unused categories do not

Table 4: Means across 48 disagreement worlds. Target regret and target success use the same source-calibrated, target-input, margin-weighted cell-policy construction after representation selection. Native PPO return evaluates the selected recurrent actor before the cellwise policy is fitted. Source return evaluates the source-weighted cell policy. Uniform random is the admitted-pool mean. Target oracle uses target truth only after lock.

Selector	Target regret	Target success	Native PPO return	Source return
KBT	**0.091498**	**0.541670**	**0.541673**	0.533782
Raw impurity selector	0.099838	0.528626	0.528907	0.528766
Source K_ρ selector	0.105021	0.528663	0.530990	**0.539506**
Source validation selector	0.107763	0.527948	0.528606	0.538990
Uniform random	0.116368	0.520698	0.521019	0.527235
Target oracle	0.090058	0.542141	0.541385	0.533318

explain the selector difference.

These results give K_ρ a practical role in partially observable RL. A system can train several recurrent encoders or retain several checkpoints, collect unlabelled histories from a deployment distribution, estimate decision margins from source calibration, and select the representation that minimises estimated irreducible regret before target rewards arrive. Source validation ranks source performance. Raw impurity treats every decision conflict alike. Equation (2) instead measures which distinctions the learned memory has merged and weights each merger by its decision consequence. Lower K_ρ then improves both the representation floor and, in these experiments, the performance of the selected cell policy and native PPO actor.

The two designs answer different questions. Fixed worlds estimate average deployment benefit and retain agreement as a genuine zero difference. Disagreement worlds estimate which selector chooses better when their pre-target recommendations differ robustly. The latter result is conditional on disagreement and does not estimate how often deployment changes. Screening changed no target outcome because target classes, rewards, posteriors, and fresh target K_ρ were unavailable until all source stages had locked.

The scope remains controlled. This is a custom delayed-cue POMDP rather than a broad control benchmark. The six categorical states are a quotient of a 32-unit recurrent state, not a six-unit neural memory. Answers do not change subsequent observations, and K_ρ is used after PPO training rather than as an end-to-end PPO loss. Earlier exact-identity, discretisation, architecture, and training-diagnostic experiments are reported in the electronic supplementary material. Action-dependent control on a memory benchmark such as POPGym or POBAX and end-to-end optimisation of a differentiable estimator remain open.

7 Conclusion

At a fixed representation-test quotient, minimum average normalised regret is margin-weighted forgetting. For an actual policy, total regret decomposes into irreducible aliasing cost and avoidable within-cell misreporting. On the lifted diagnostic task, the same quantity is a weighted freedom deficit. The lift removes cross-cell extension structure, so the deficit reduces to cell-partition Bayes error. In a general embodied language, freedom retains the extension structure and need not reduce to Bayes error.

The identity extends regret minimisation to representation search and yields a diagnostic-

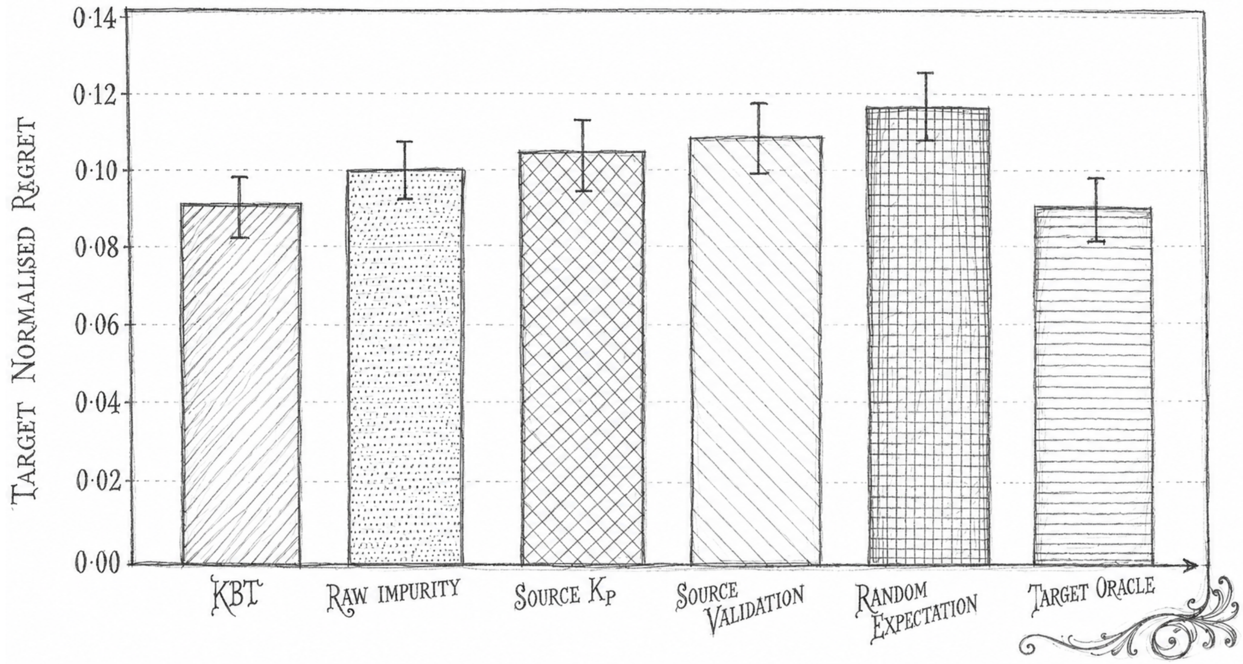

Figure 2: Mean target normalised regret across 48 disagreement worlds. Error bars are 95% bootstrap intervals over worlds. Lower is better.

quotient convergence theorem, a regret preorder on abstractions, stability under margin estimation error, and a multi-class generalisation. Under the canonical independent prior, the bridge converts the regret floor into generalisation probability $e^{-K_\rho(M)}$. External analysis shows that expected free energy can follow the freedom order only under specified embodied outcomes, forecasts, and preferences. Changing forecasts or preferences can reverse the order, so expected free energy does not independently determine freedom [14].

The recurrent PPO experiment turns the representation-search theorem into a model-selection procedure. Across 48 fixed worlds and 2,880 candidate representations, a source-calibrated estimate of K_ρ evaluated on unlabelled target histories ranked candidates better than raw impurity and reduced rank-one fresh target K_ρ. Across two further 24-world replications where the selectors made different fold-stable choices, it reduced fresh target K_ρ by 6.3%, reduced normalised regret by 8.4%, and increased target success by 1.30 percentage points. KBT selected lower-K_ρ memory in 43 of 48 disagreement worlds and increased success in 47, with one tie. Selector identities and cell policies were locked before target classes, correct answers, rewards, or regret were revealed.

This can improve RL systems through an outer representation-selection loop. Train candidate recurrent or world-model encoders on source interaction, retain checkpoints or fixed-budget quotients, collect unlabelled deployment histories, and choose the candidate with least estimated K_ρ. Source validation rewards source performance. Predictive losses reward every prediction error according to their training objective. Raw impurity treats every decision conflict alike. K_ρ targets the part no policy reading the selected representation can repair, weighted by how much each merged distinction changes attainable value.

The experiment isolates this mechanism. Answers do not change later observations, and the cellwise policy is fitted after PPO training. Native PPO actors selected by KBT also had higher mean target return than those selected by raw impurity in the fixed and disagreement cohorts. The

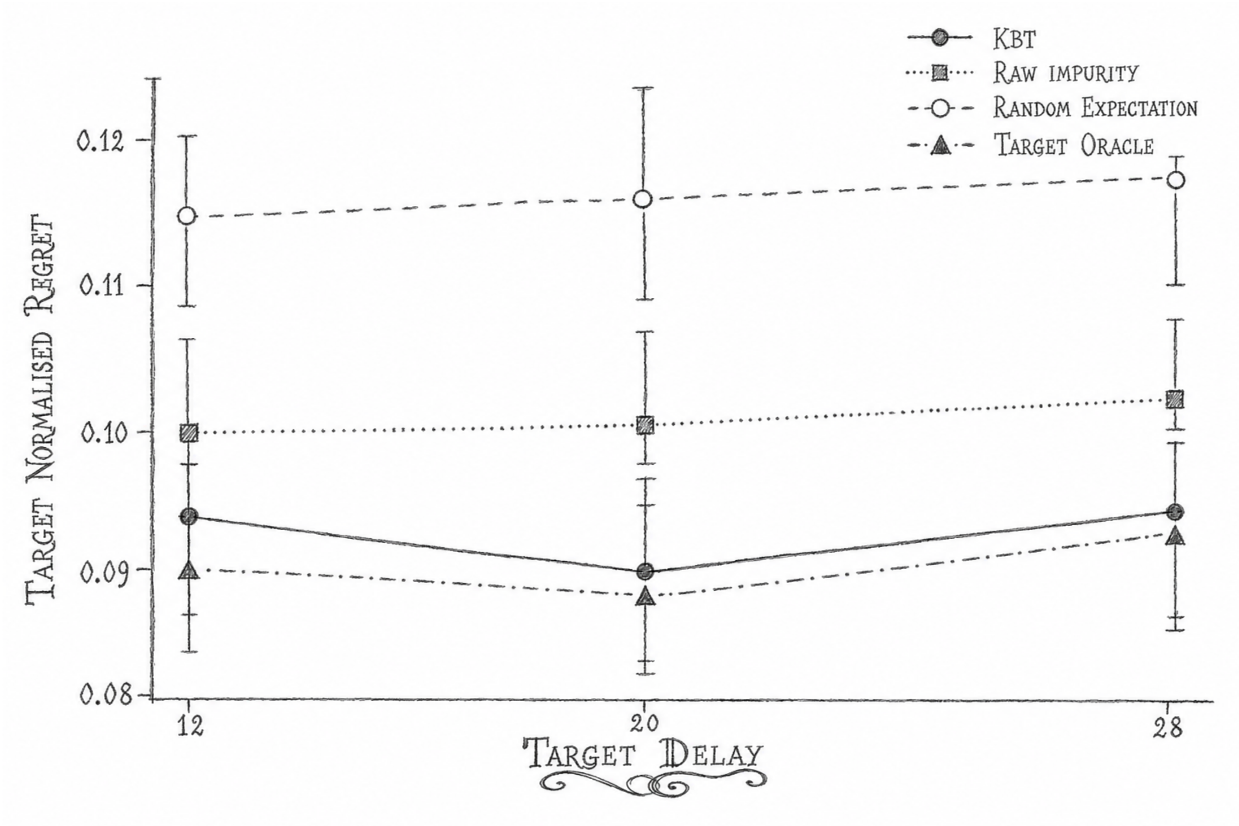

Figure 3: Target regret by delay across 48 disagreement worlds. Error bars are 95% bootstrap intervals over worlds. KBT reduces regret relative to raw impurity by 0.006842 at delay 12, 0.009957 at delay 20, and 0.008294 at delay 28.

next test is action-dependent control on a memory-improvable benchmark, followed by end-to-end optimisation of a differentiable estimator. Independent neural work on joint completion remains relevant off the lifted task, where one shared function class couples decisions across cells [12].

Data availability

The electronic supplementary material contains the frozen plan, source code, target-input and lock records, fixed-world candidate tables, disagreement screening records, retained-world results, quantiser artefacts, selector records, statistical recomputation, audit outputs, and earlier diagnostic experiments. The compact review archive omits deterministic PyTorch states and generated input arrays, but regenerates them from the frozen seeds and settings. The numerical summaries and figures in the article can be recomputed from the included tables.

Competing interests

I declare no competing interests.

Funding

No external funding supported this study.

References

[1] David Abel, D. Ellis Hershkowitz, and Michael L. Littman. Near optimal behavior via approximate state abstraction. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2915–2923. PMLR, 2016.

[2] David Abel, Dilip Arumugam, Lucas Lehnert, and Michael L. Littman. State abstractions for lifelong reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 10–19. PMLR, 2018.

[3] Elias Bareinboim, Juan D. Correa, Duligur Ibeling, and Thomas Icard. On Pearl's hierarchy and the foundations of causal inference. In *Probabilistic and Causal Inference: The Works of Judea Pearl*, pages 507–556. ACM, 2022. doi: 10.1145/3501714.3501743.

[4] Yoshua Bengio, Aaron Courville, and Pascal Vincent. Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8): 1798–1828, 2013.

[5] Yoshua Bengio, Tristan Deleu, Nasim Rahaman, Nan Rosemary Ke, Sebastien Lachapelle, Olexa Bilaniuk, Anirudh Goyal, and Christopher Pal. A meta-transfer objective for learning to disentangle causal mechanisms. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=ryxWIgBFPS.

[6] Michael Timothy Bennett. The optimal choice of hypothesis is the weakest, not the shortest. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 42–51. Springer, 2023. doi: 10.1007/978-3-031-33469-6_5. URL https://doi.org/10.1007/978-3-031-33469-6_5.

[7] Michael Timothy Bennett. Emergent causality and the foundation of consciousness. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 52–61. Springer, 2023. doi: 10.1007/978-3-031-33469-6_6. URL https://doi.org/10.1007/978-3-031-33469-6_6.

[8] Michael Timothy Bennett. Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer, 2024. doi: 10.1007/978-3-031-65572-2_3. URL https://doi.org/10.1007/978-3-031-65572-2_3.

[9] Michael Timothy Bennett. Are biological systems more intelligent than artificial intelligence?, 2024. URL https://arxiv.org/abs/2405.02325. In press, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds.

[10] Michael Timothy Bennett. Optimal policy is weakest policy. In *Artificial General Intelligence*, volume 16057 of *Lecture Notes in Computer Science*, pages 43–56. Springer, 2025. doi: 10.1007/978-3-032-00686-8_5. URL https://doi.org/10.1007/978-3-032-00686-8_5.

[11] Michael Timothy Bennett. *How To Build Conscious Machines*. PhD thesis, The Australian National University, 2025. URL `https://hdl.handle.net/1885/733782452`.

[12] Michael Timothy Bennett. Are flat minima an illusion?, 2026. URL `https://arxiv.org/abs/2605.05209`.

[13] Michael Timothy Bennett. The wrong razor. In *Proceedings of the 19th International Conference on Artificial General Intelligence*. Springer Nature, 2026.

[14] Michael Timothy Bennett. Free energy cannot define intelligence, 2026.

[15] Michael Timothy Bennett. *How to Build Conscious Machines: Deriving Intelligence, Life and Mind from Time*. Palgrave Macmillan, 2027. Forthcoming.

[16] Rodney A. Brooks. Intelligence without representation. *Artificial Intelligence*, 47(1-3):139–159, 1991. doi: 10.1016/0004-3702(91)90053-m. URL `https://doi.org/10.1016/0004-3702(91)90053-m`.

[17] Rosa Cao and Daniel Yamins. Explanatory models in neuroscience, part 2: Functional intelligibility and the contravariance principle. *Cognitive Systems Research*, 85:101200, 2024.

[18] Pablo Samuel Castro. Scalable methods for computing state similarity in deterministic Markov decision processes. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(06): 10069–10076, 2020.

[19] Norman Ferns, Prakash Panangaden, and Doina Precup. Metrics for finite Markov decision processes. In *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence*, pages 162–169. AUAI Press, 2004.

[20] Norman Ferns, Prakash Panangaden, and Doina Precup. Bisimulation metrics for continuous Markov decision processes. *SIAM Journal on Computing*, 40(6):1662–1714, 2011. doi: 10.1137/10080484X.

[21] Carles Gelada, Saurabh Kumar, Jacob Buckman, Ofir Nachum, and Marc G. Bellemare. DeepMDP: Learning continuous latent space models for representation learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2170–2179. PMLR, 2019.

[22] Robert Givan, Thomas Dean, and Matthew Greig. Equivalence notions and model minimization in Markov decision processes. *Artificial Intelligence*, 147(1-2):163–223, 2003.

[23] Anirudh Goyal and Yoshua Bengio. Inductive biases for deep learning of higher-level cognition. *Proceedings of the Royal Society A*, 478(2266):20210068, 2022.

[24] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640(8059):647–653, 2025. doi: 10.1038/s41586-025-08744-2.

[25] Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *The Twelfth International Conference on Learning Representations*, 2024.

[26] Jinu Hyeon, Woobin Park, Hongjoon Ahn, and Taesup Moon. Action-sufficient goal representations, 2026. URL `https://arxiv.org/abs/2601.22496`.

[27] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1-2):99–134, 1998.

[28] Nan Rosemary Ke, Olexa Bilaniuk, Anirudh Goyal, Stefan Bauer, Hugo Larochelle, Bernhard Schölkopf, Michael C. Mozer, Chris Pal, and Yoshua Bengio. Systematic evaluation of causal discovery in visual model based reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[29] Finnian Lattimore, Tor Lattimore, and Mark D. Reid. Causal bandits: Learning good interventions via causal inference. In *Advances in Neural Information Processing Systems*, volume 29, pages 1181–1189, 2016.

[30] Lihong Li, Thomas J. Walsh, and Michael L. Littman. Towards a unified theory of state abstraction for MDPs. In *International Symposium on Artificial Intelligence and Mathematics*, 2006.

[31] David Marr. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W.H. Freeman, 1982.

[32] Patrick McMillen and Michael Levin. Collective intelligence: A unifying concept for integrating biology across scales and substrates. *Communications Biology*, 7(1):378, March 2024. doi: 10.1038/s42003-024-06037-4. URL `https://doi.org/10.1038/s42003-024-06037-4`.

[33] Tom M. Mitchell. Generalization as search. *Artificial Intelligence*, 18(2):203–226, 1982. doi: 10.1016/0004-3702(82)90040-6.

[34] Tom M. Mitchell. *Machine Learning*. McGraw-Hill, 1997.

[35] Steven Morad, Ryan Kortvelesy, Matteo Bettini, Stephan Liwicki, and Amanda Prorok. POP-Gym: Benchmarking partially observable reinforcement learning. In *International Conference on Learning Representations*, 2023.

[36] Aran Nayebi. What capable agents must know: Selection theorems for robust decision-making under uncertainty. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence*, volume 337 of *Proceedings of Machine Learning Research*, pages 4773–4795. PMLR, 2026. URL `https://proceedings.mlr.press/v337/nayebi26a.html`.

[37] Tianwei Ni, Benjamin Eysenbach, and Ruslan Salakhutdinov. Recurrent model-free RL can be a strong baseline for many POMDPs. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 16691–16723. PMLR, 2022. URL `https://proceedings.mlr.press/v162/ni22a.html`.

[38] Tianwei Ni, Michel Ma, Benjamin Eysenbach, and Pierre-Luc Bacon. When do transformers shine in RL? decoupling memory from credit assignment. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[39] Tianwei Ni, Benjamin Eysenbach, Erfan Seyedsalehi, Michel Ma, Clement Gehring, Aditya Mahajan, and Pierre-Luc Bacon. Bridging state and history representations: Understanding self-predictive RL. In *The Twelfth International Conference on Learning Representations*, 2024.

[40] Judea Pearl. Causal diagrams for empirical research. *Biometrika*, 82(4):669–688, 1995.

[41] Judea Pearl. *Causality: Models, Reasoning, and Inference.* Cambridge University Press, Cambridge, 2 edition, 2009.

[42] M. O. Rabin and D. Scott. Finite automata and their decision problems. *IBM Journal of Research and Development*, 3(2):114–125, 1959. doi: 10.1147/rd.32.0114. URL https://doi.org/10.1147/rd.32.0114.

[43] Jonathan Richens and Tom Everitt. Robust agents learn causal world models. In *The Twelfth International Conference on Learning Representations*, 2024. doi: 10.48550/arxiv.2402.10877. URL https://arxiv.org/abs/2402.10877.

[44] Jonathan Richens, Tom Everitt, and David Abel. General agents need world models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 51659–51687. PMLR, 2025. URL https://proceedings.mlr.press/v267/richens25a.html.

[45] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.

[46] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. doi: 10.48550/arXiv.1707.06347. URL https://arxiv.org/abs/1707.06347.

[47] Ricard Solé, Luis F. Seoane, Jordi Pla-Mauri, Michael Timothy Bennett, Michael E. Hochberg, and Michael Levin. Cognition spaces: natural, artificial, and hybrid, 2026. URL https://arxiv.org/abs/2601.12837.

[48] Ricard Solé, Melanie Moses, and Stephanie Forrest. Liquid brains, solid brains. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 374(1774):20190040, 2019. doi: 10.1098/rstb.2019.0040. URL https://royalsocietypublishing.org/doi/abs/10.1098/rstb.2019.0040.

[49] Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search.* MIT Press, Cambridge, MA, 2 edition, 2000.

[50] Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction.* MIT press, MA, 2018.

[51] Ruo Yu Tao, Kaicheng Guo, Cameron Allen, and George Konidaris. Benchmarking partial observability in reinforcement learning with a suite of memory-improvable domains. In *Proceedings of the Second Reinforcement Learning Conference*, 2025.

[52] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000.

[53] D.H. Wolpert and W.G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997. doi: 10.1109/4235.585893. URL https://doi.org/10.1109/4235.585893.

[54] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks? In *Proceedings of the 28th International Conference on Neural*

Information Processing Systems - Volume 2, NIPS'14, page 3320–3328, Cambridge, MA, USA, 2014. MIT Press.

[55] Amy Zhang, Rowan McAllister, Roberto Calandra, Yarin Gal, and Sergey Levine. Learning invariant representations for reinforcement learning without reconstruction. In *International Conference on Learning Representations*, 2021.